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(54) **RESERVOIR CONFIGURATION FOR
AEROSOL DELIVERY DEVICE**

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22, 2019.

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(2020.01); **A24F 40/42** (2020.01); **A24F 40/44**
(2020.01); **B05B 17/0684** (2013.01)

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40/05; A24F 40/10; A24F 40/42; A24F
40/44

See application file for complete search history.

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Primary Examiner — Philip Y Louie

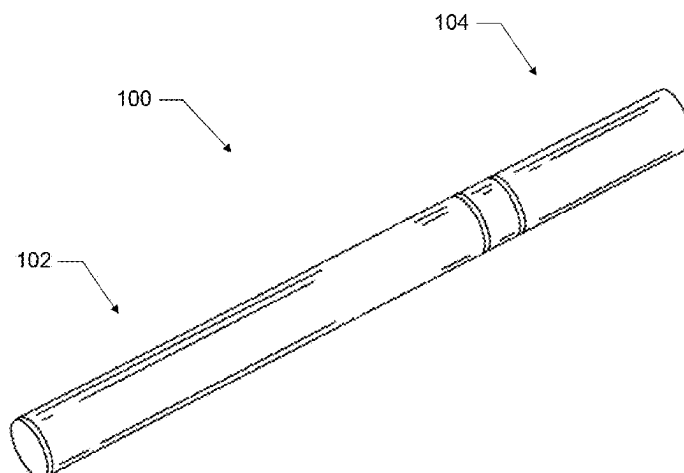
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(57) **ABSTRACT**

The present disclosure provides an aerosol delivery device that may comprise a housing defining an outer wall. The device may further include a power source and a control component, a mouthpiece portion, a tank portion that includes a reservoir configured to contain a liquid composition, and an atomization assembly configured to vaporize the liquid composition to generate an aerosol. The atomization assembly may comprise a vibrating assembly that includes a mesh plate. In some implementations, the reservoir of the aerosol delivery device may be configured to rotate relative to the position of the aerosol delivery device. In some implementations, the aerosol delivery device may further comprise a perforated gate. In some implementations, the aerosol delivery device may further comprise a liquid transport element. In some implementations, the aerosol delivery device may further comprise a micropump assembly. In some implementations, the reservoir of the aerosol delivery device may be U-shaped.

5 Claims, 17 Drawing Sheets



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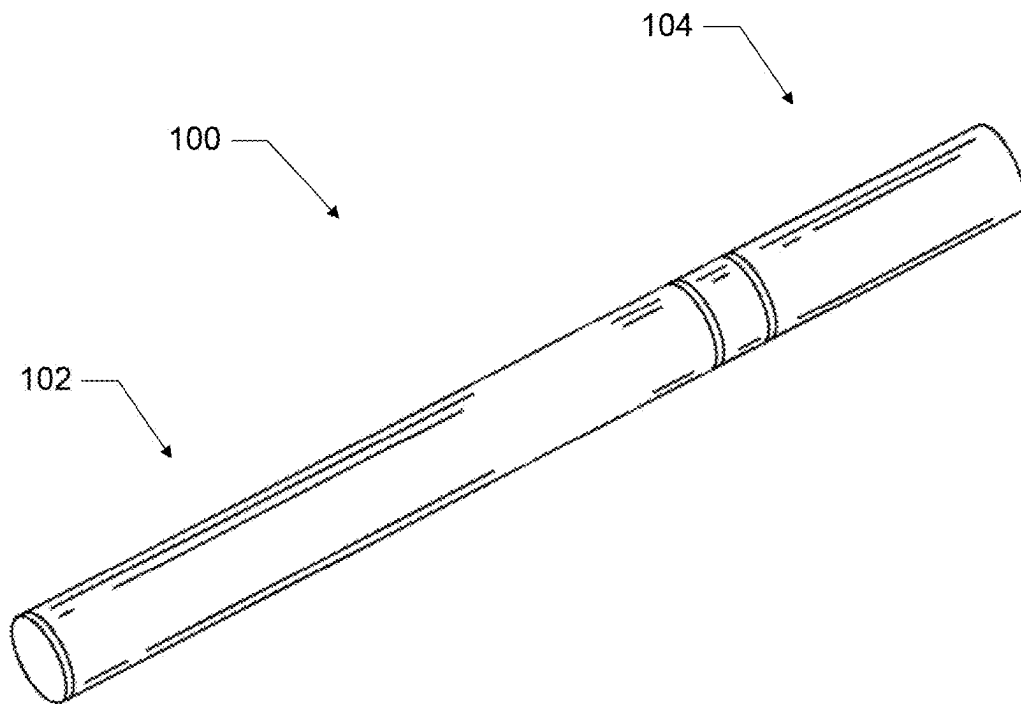


FIG. 1

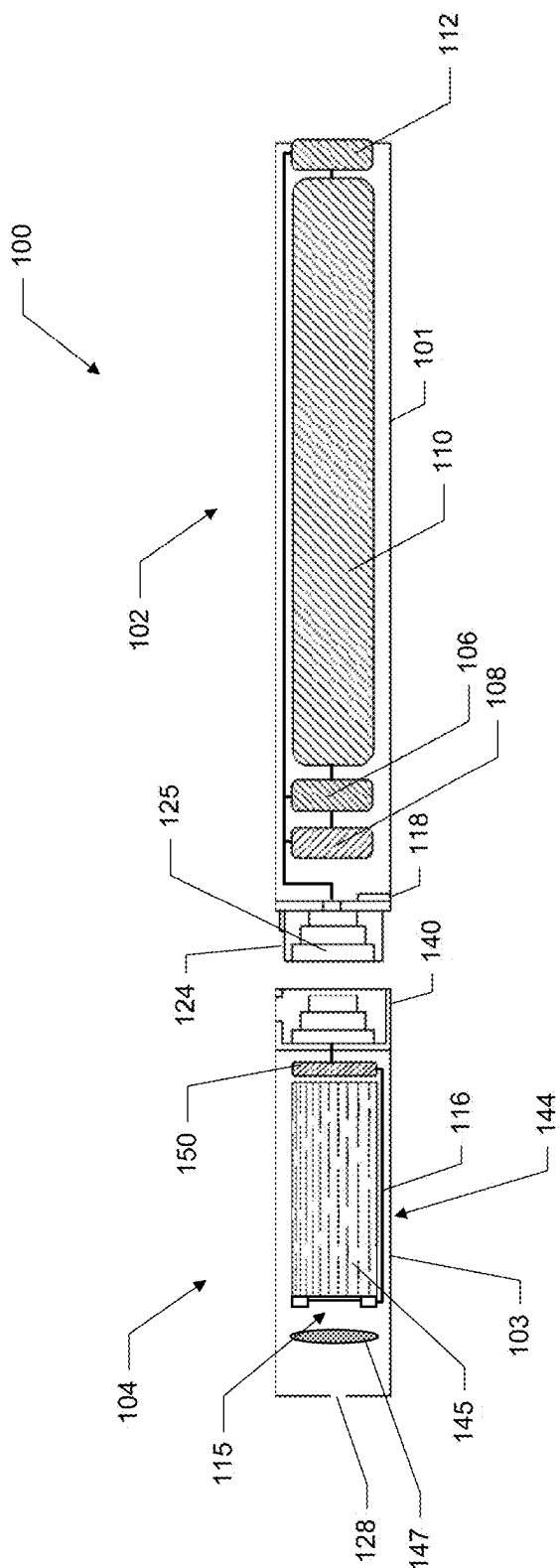


FIG. 2

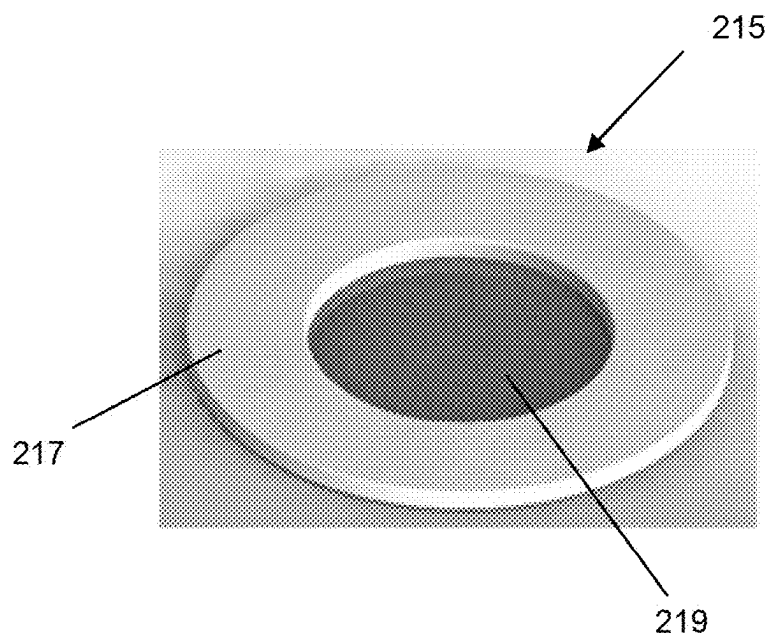


FIG. 3

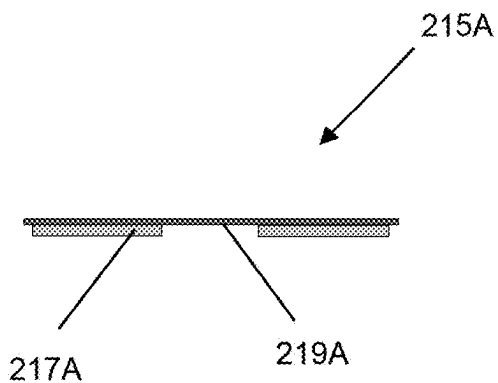


FIG. 4A

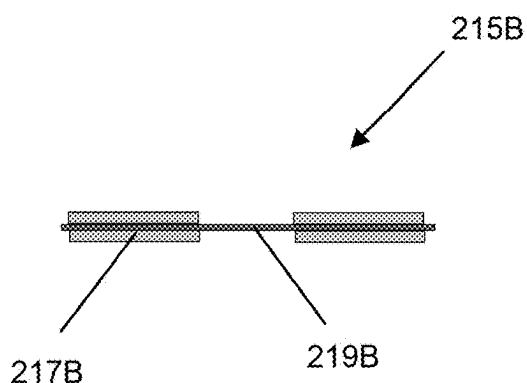


FIG. 4B

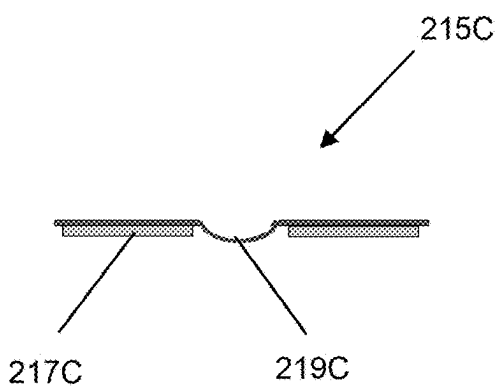


FIG. 4C

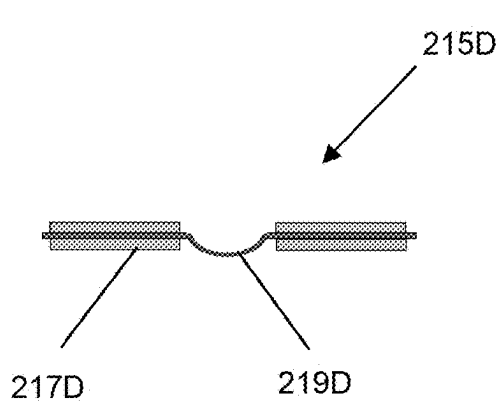


FIG. 4D

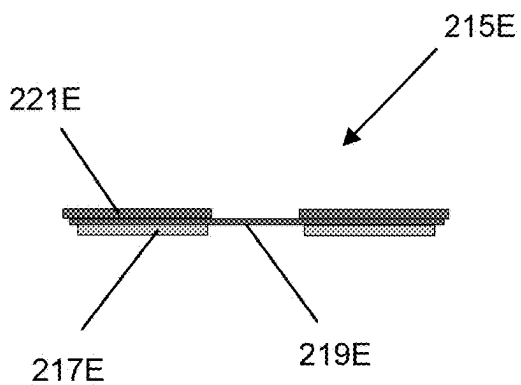


FIG. 4E

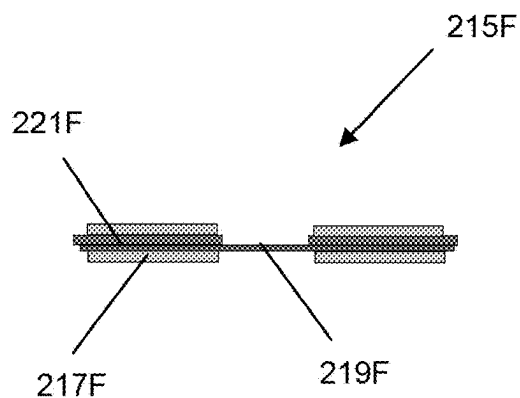


FIG. 4F

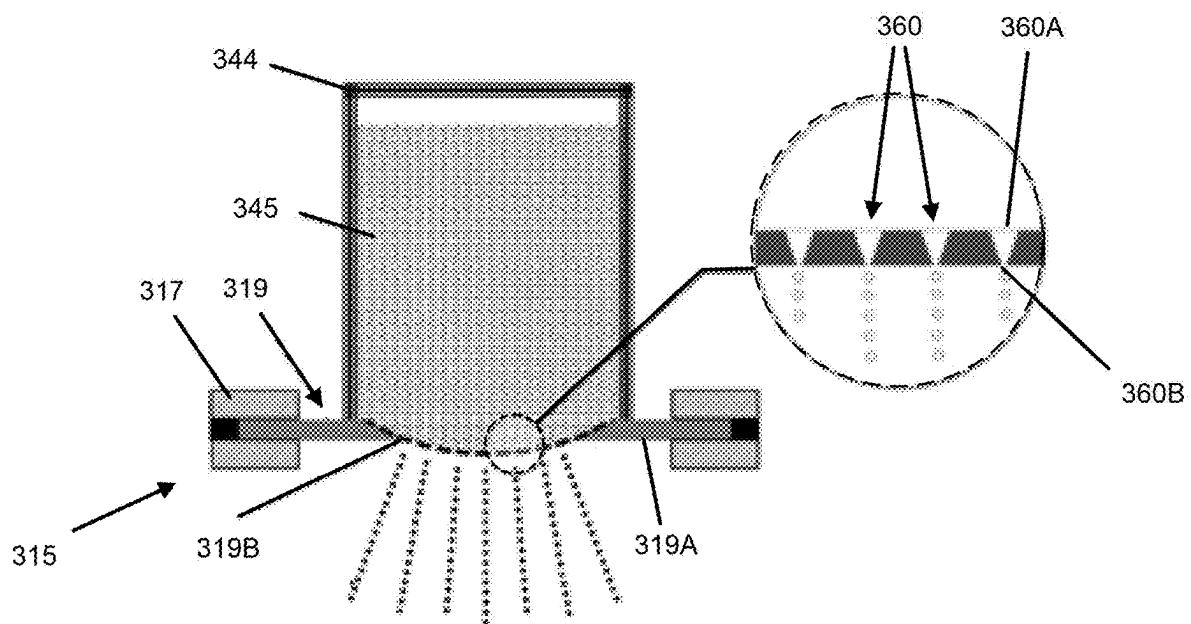


FIG. 5

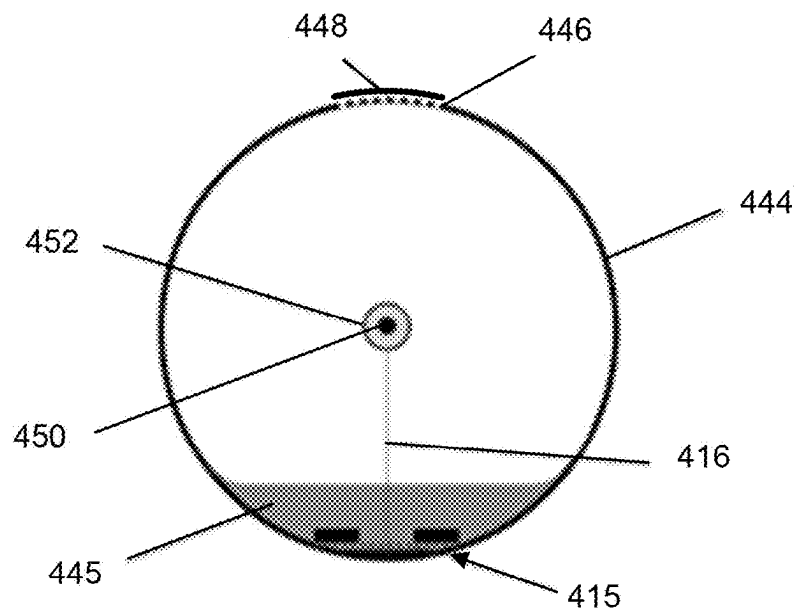


FIG. 6

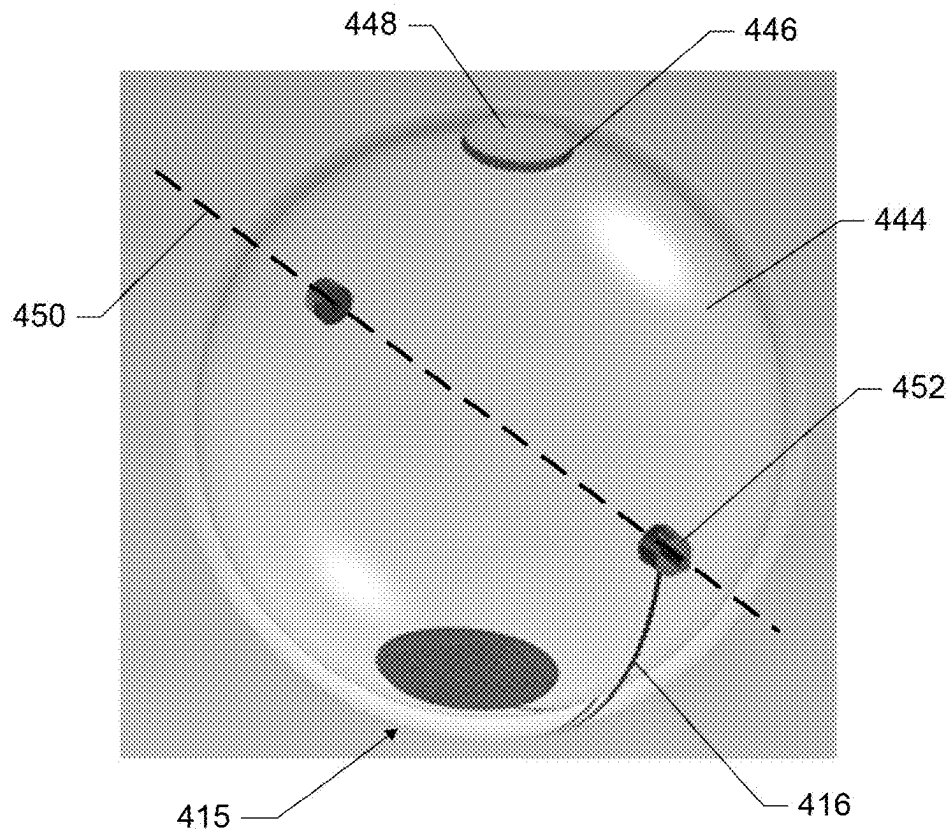


FIG. 7

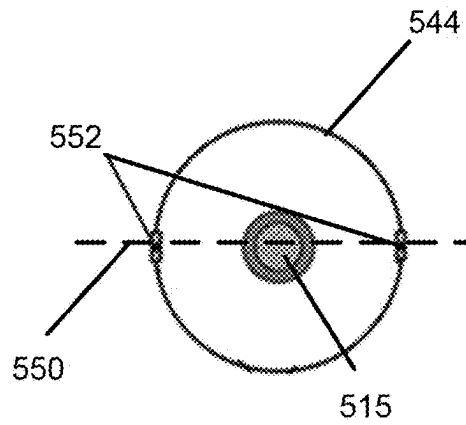


FIG. 8A

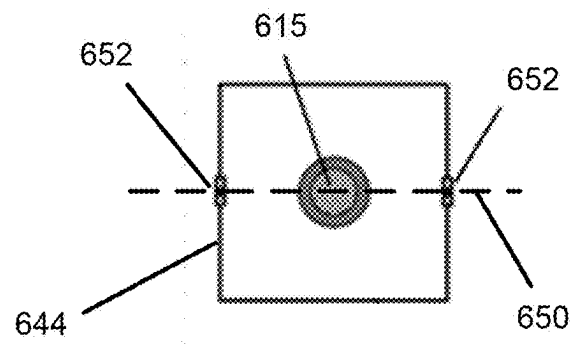


FIG. 8B

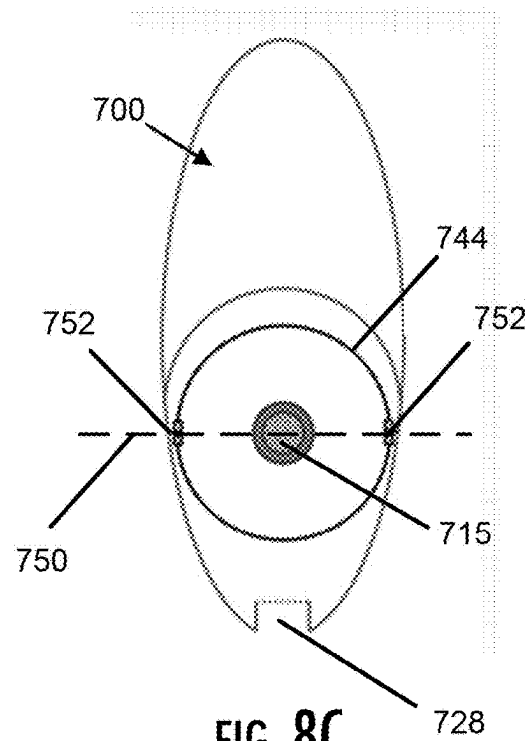


FIG. 8C

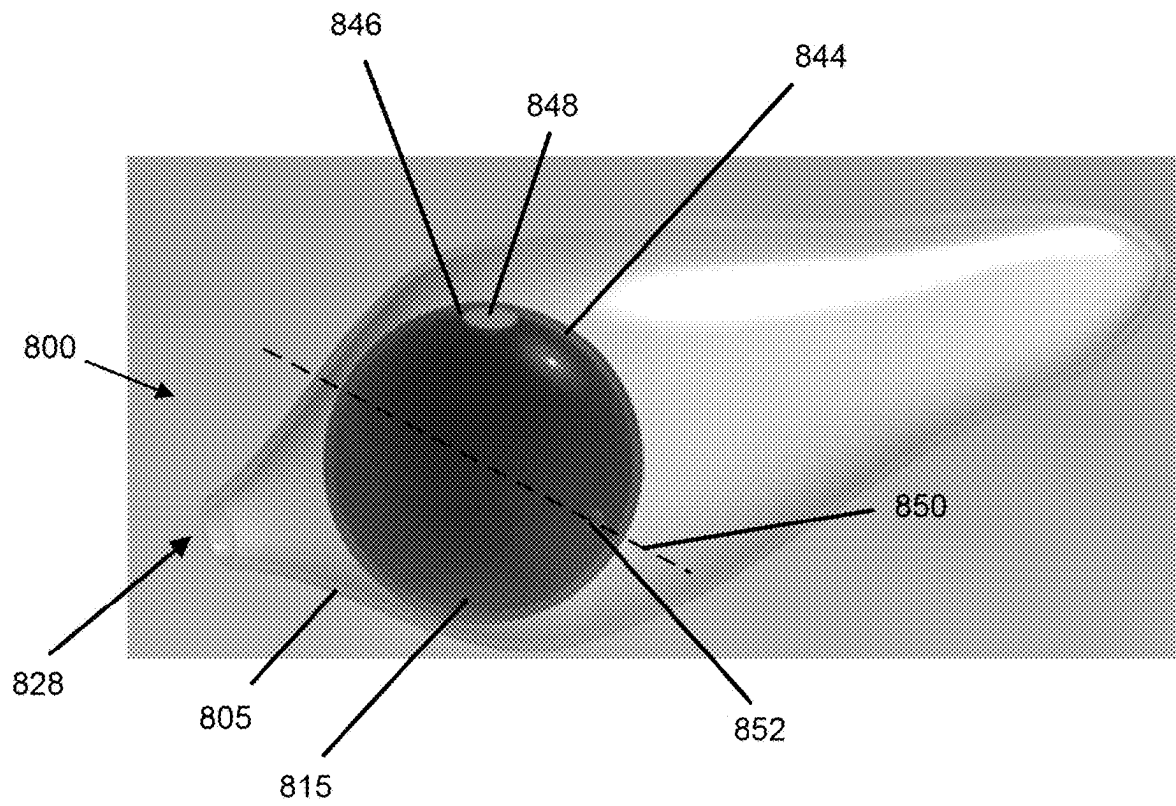


FIG. 9

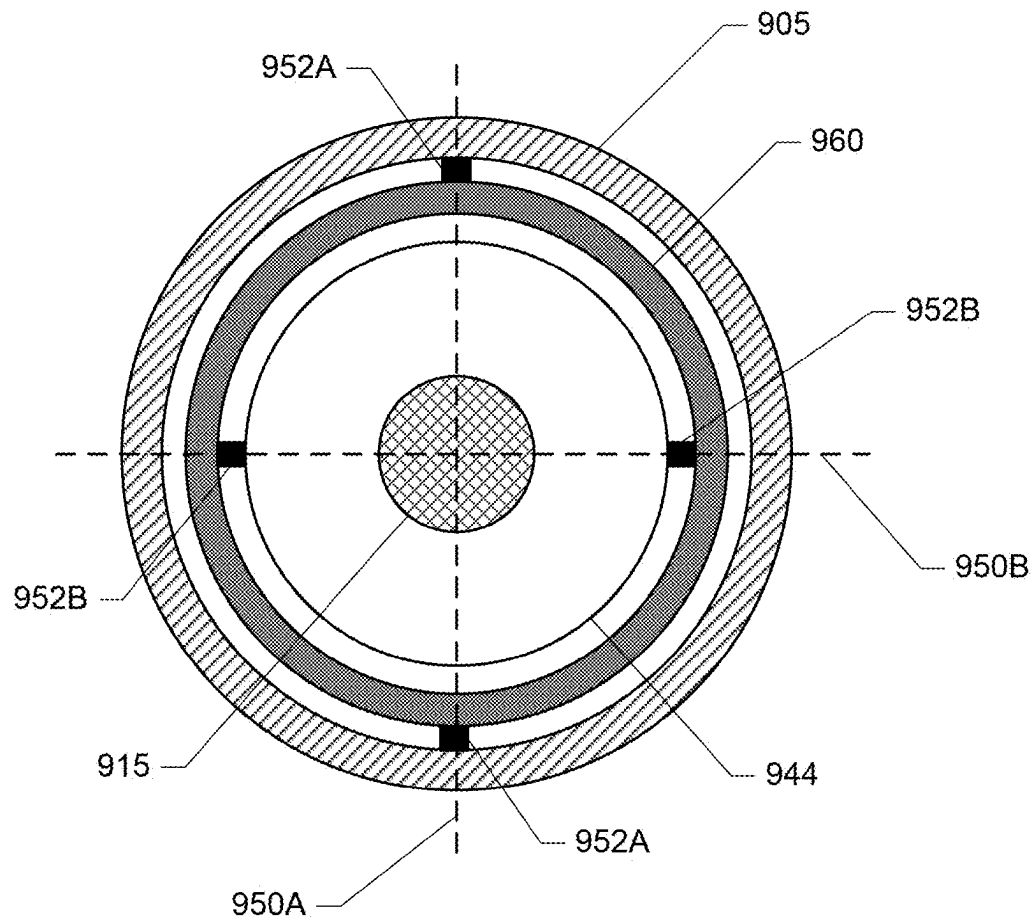


FIG. 10

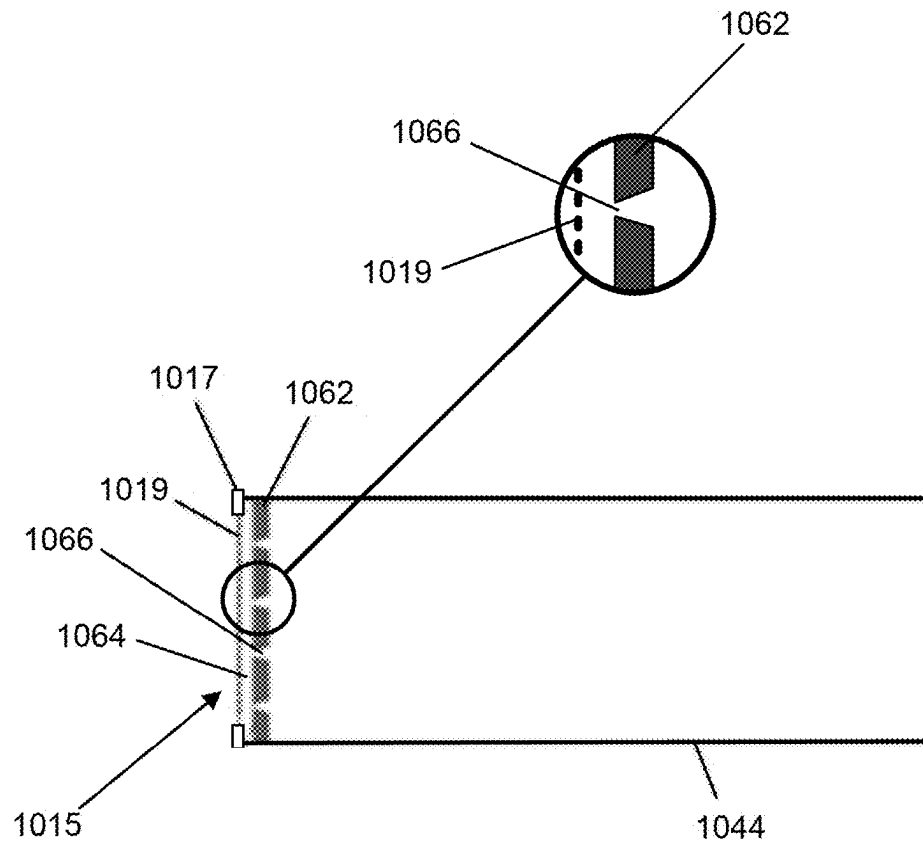


FIG. 11

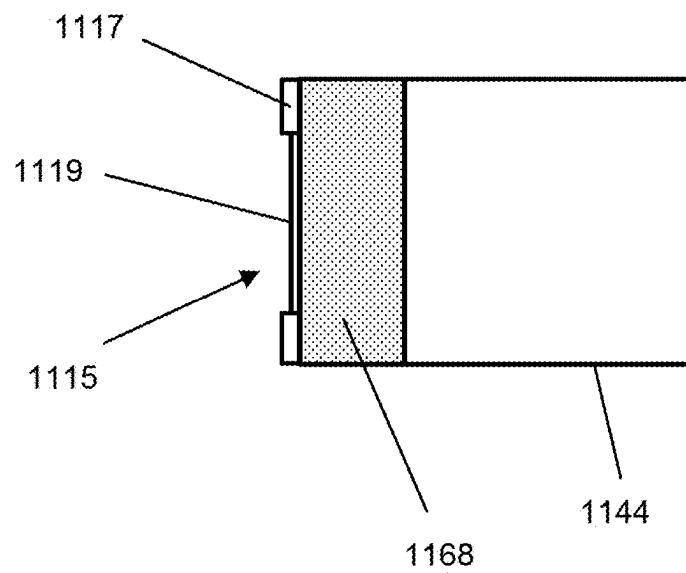


FIG. 12

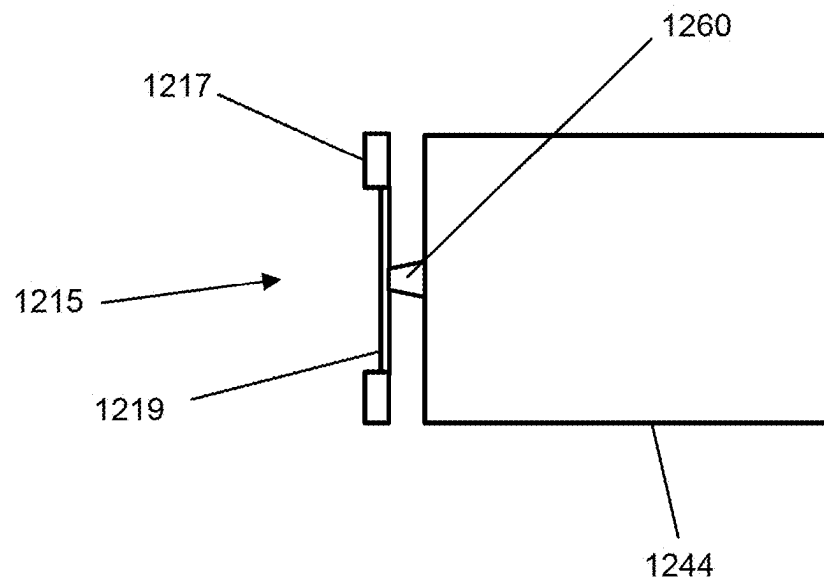


FIG. 13

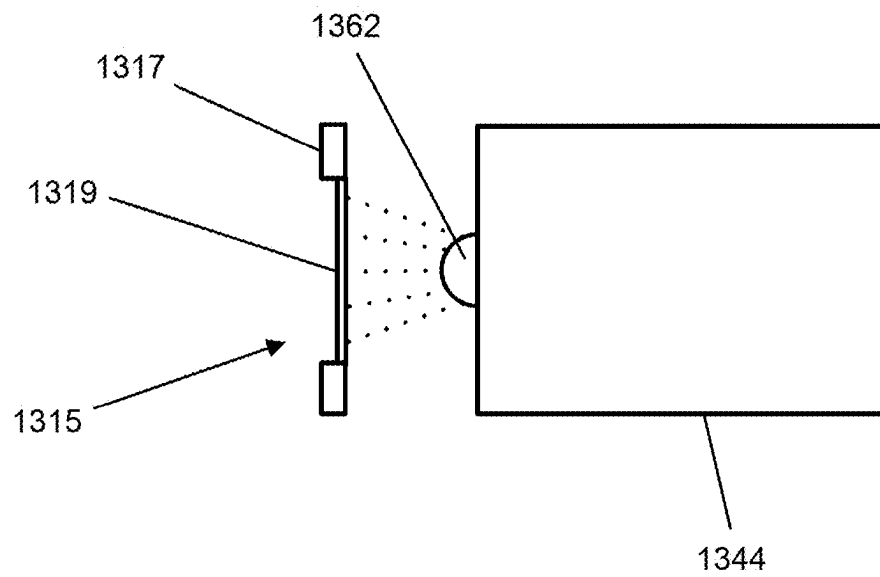
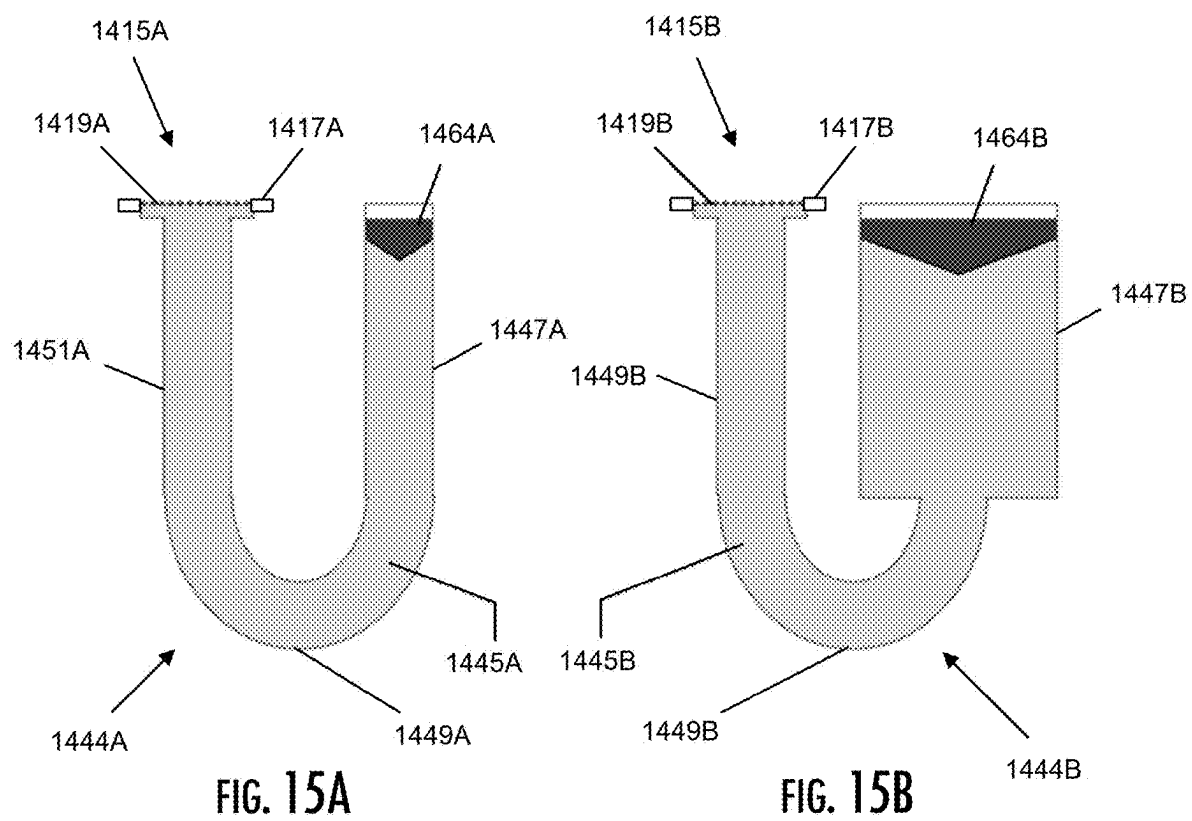


FIG. 14



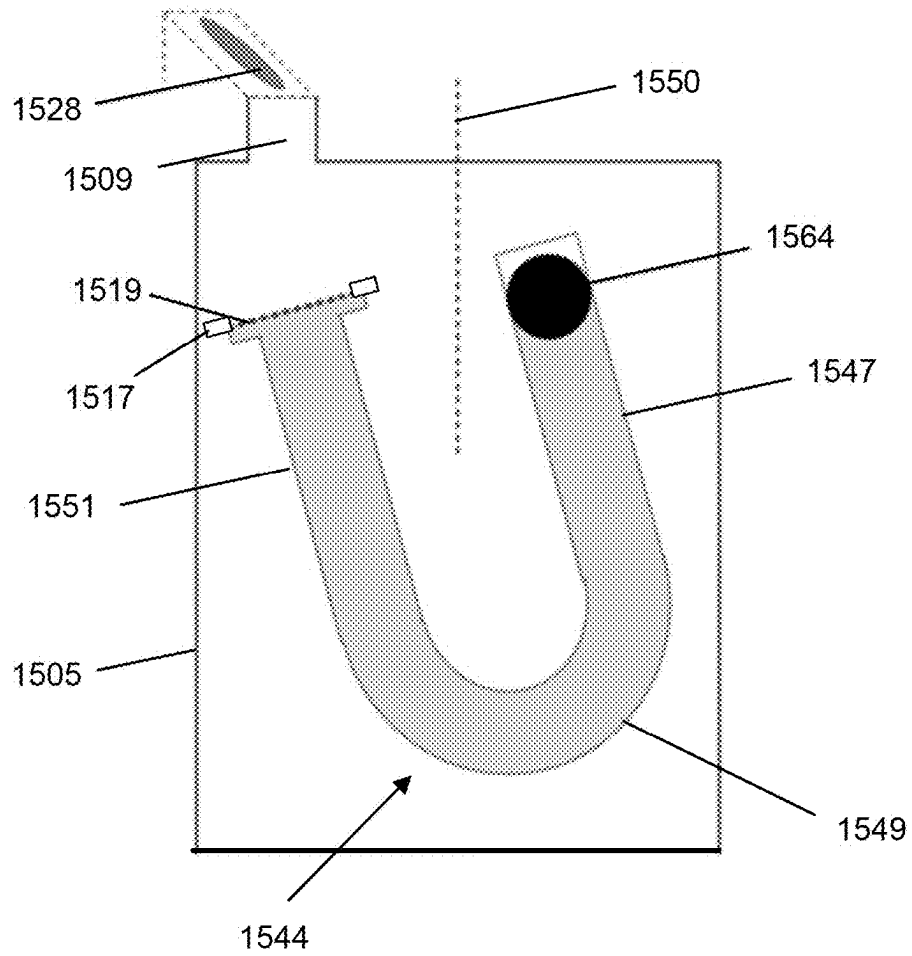


FIG. 16

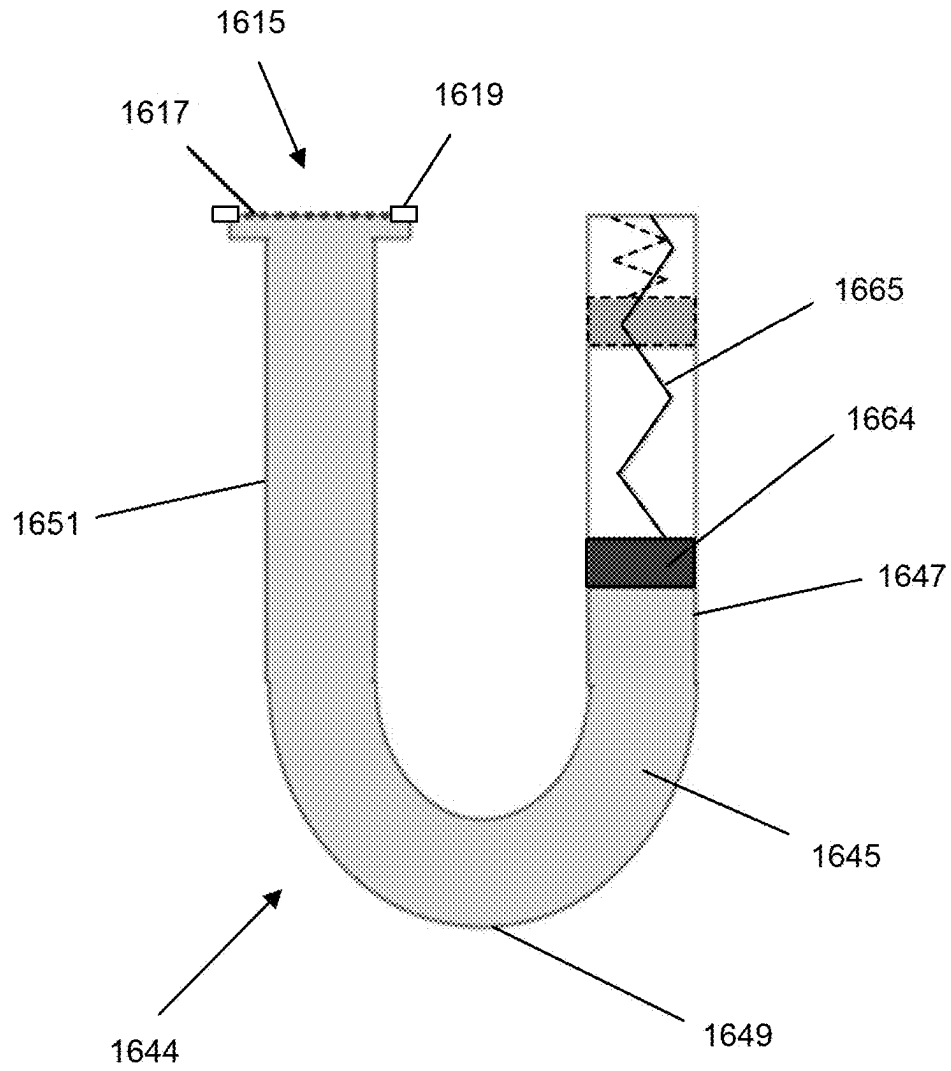


FIG. 17

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RESERVOIR CONFIGURATION FOR AEROSOL DELIVERY DEVICE

CROSS-REFERENCE TO RELATED APPLICATIONS

This application claims priority to, and the benefit of, U.S. Provisional Patent Application No. 62/851,318, titled Reservoir Configuration for Aerosol Delivery Device, filed on May 22, 2019, which is incorporated herein in its entirety by reference.

TECHNOLOGICAL FIELD

The present disclosure relates to aerosol delivery devices, and more particularly to an aerosol delivery device that includes a reservoir and an atomization assembly, which may utilize electrical power to vaporize an aerosol precursor composition for the production of an aerosol. In various implementations, the aerosol precursor composition, which may incorporate materials and/or components that may be made or derived from tobacco or otherwise incorporate tobacco or other plants, may include natural or synthetic components including flavorants, and/or may include one or more medicinal components, is vaporized by the atomization assembly to produce an inhalable substance for human consumption.

BACKGROUND

Many smoking devices have been proposed through the years as improvements upon, or alternatives to, smoking products that require combusting tobacco for use. Many of those devices purportedly have been designed to provide the sensations associated with cigarette, cigar, or pipe smoking, but without delivering considerable quantities of incomplete combustion and pyrolysis products that result from the burning of tobacco. To this end, there have been proposed numerous smoking products, flavor generators, and medicinal inhalers that utilize electrical energy to vaporize or heat a volatile material, or attempt to provide the sensations of cigarette, cigar, or pipe smoking without burning tobacco to a significant degree. See, for example, the various alternative smoking articles, aerosol delivery devices, and heat generating sources set forth in the background art described in U.S. Pat. No. 7,726,320 to Robinson et al., U.S. Pat. App. Pub. No. 2013/0255702 to Griffith Jr. et al., and U.S. Pat. App. Pub. No. 2014/0096781 to Sears et al., which are incorporated herein by reference in their entireties. See also, for example, the various types of smoking articles, aerosol delivery devices, and electrically powered sources referenced by brand name and commercial source in U.S. Pat. App. Pub. No. 2015/0216232 to Bless et al., which is incorporated herein by reference in its entirety.

However, it would be desirable to provide an aerosol delivery device with enhanced functionality. In this regard, it is desirable to provide an aerosol delivery with advantageous features.

BRIEF SUMMARY

The present disclosure relates to aerosol delivery devices, methods of forming such devices, and elements of such devices. The present disclosure includes, without limitation, the following example implementations.

An aerosol delivery device comprising a housing defining an outer wall, and further including a power source and a

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control component, a mouthpiece portion, a tank portion that includes a reservoir configured to contain a liquid composition, and an atomization assembly configured to vaporize the liquid composition to generate an aerosol, wherein the atomization assembly comprises a vibrating assembly that includes a mesh plate, and the reservoir is configured to rotate relative to the position of the aerosol delivery device.

The aerosol delivery device of any preceding example implementation, or any combination of any preceding example implementations, wherein the mesh plate is substantially flat.

The aerosol delivery device of any preceding example implementation, or any combination of any preceding example implementations, wherein at least a portion of the mesh plate is convex with respect to the reservoir.

The aerosol delivery device of any preceding example implementation, or any combination of any preceding example implementations, wherein the vibrating assembly further includes a piezoelectric component affixed to and substantially surrounding the mesh plate.

The aerosol delivery device of any preceding example implementation, or any combination of any preceding example implementations, wherein the vibrating assembly is configured to be located proximate a closed portion of the reservoir.

The aerosol delivery device of any preceding example implementation, or any combination of any preceding example implementations, wherein the vibrating assembly is located proximate an open portion of the reservoir.

The aerosol delivery device of any preceding example implementation, or any combination of any preceding example implementations, wherein the reservoir is substantially spherical.

The aerosol delivery device of any preceding example implementation, or any combination of any preceding example implementations, wherein the reservoir is substantially cylindrical.

The aerosol delivery device of any preceding example implementation, or any combination of any preceding example implementations, wherein the reservoir is configured to rotate about a single axis.

The aerosol delivery device of any preceding example implementation, or any combination of any preceding example implementations, further comprising a mid-frame configured connect the reservoir to housing, wherein the mid-frame is configured to rotate relative to the housing about a first axis, wherein the reservoir is configured to rotate relative to the mid-frame about a second axis, and wherein the first and second axes are substantially perpendicular.

An aerosol delivery device comprising a housing defining an outer wall, and further including a power source and a control component, a mouthpiece portion, a tank portion that includes a reservoir configured to contain a liquid composition, and an atomization assembly configured to vaporize the liquid composition to generate an aerosol, wherein the atomization assembly comprises a vibrating assembly that includes a mesh plate, and a perforated gate having a plurality of openings defined therein, the perforated gate located proximate the mesh plate and between the liquid composition and the mesh plate.

The aerosol delivery device of any preceding example implementation, or any combination of any preceding example implementations, wherein the perforated gate is substantially parallel to the mesh plate.

The aerosol delivery device of any preceding example implementation, or any combination of any preceding

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example implementations, further comprising a chamber defined between the mesh plate and the perforated gate.

The aerosol delivery device of any preceding example implementation, or any combination of any preceding example implementations, wherein the plurality of openings of the perforated gate have a truncated conical shape.

The aerosol delivery device of any preceding example implementation, or any combination of any preceding example implementations, wherein small ends of the plurality of openings are located closest to the mesh plate.

The aerosol delivery device of any preceding example implementation, or any combination of any preceding example implementations, at least a portion of the surface of the perforated gate is coated with a coating.

The aerosol delivery device of any preceding example implementation, or any combination of any preceding example implementations, wherein the coating comprises a hydrophobic/oleophobic coating.

The aerosol delivery device of any preceding example implementation, or any combination of any preceding example implementations, wherein the vibrating assembly further includes a piezoelectric component affixed to and substantially surrounding the mesh plate.

The aerosol delivery device of any preceding example implementation, or any combination of any preceding example implementations, wherein the mesh plate is substantially flat.

The aerosol delivery device of any preceding example implementation, or any combination of any preceding example implementations, wherein at least a portion of the mesh plate is convex with respect to the reservoir.

An aerosol delivery device comprising a housing defining an outer wall, and further including a power source and a control component, a mouthpiece portion, a tank portion that includes a reservoir configured to contain a liquid composition, and an atomization assembly configured to vaporize the liquid composition to generate an aerosol, wherein the atomization assembly comprises a vibrating assembly that includes a mesh plate, and further includes a liquid transport element, one end of which is located proximate the mesh plate.

The aerosol delivery device of any preceding example implementation, or any combination of any preceding example implementations, wherein the liquid transport element comprises a single layer of a single material.

The aerosol delivery device of any preceding example implementation, or any combination of any preceding example implementations, wherein the liquid transport element comprises multiple layers of a single material.

The aerosol delivery device of any preceding example implementation, or any combination of any preceding example implementations, wherein the liquid transport element comprises multiple layers.

The aerosol delivery device of any preceding example implementation, or any combination of any preceding example implementations, wherein the liquid transport element comprises at least one of a polymer material, a polymer fiber material, a cotton material, a silk material, a silica fiber material, a particulate material, a synthetic fiber material, a natural fiber material, and a ceramic material.

The aerosol delivery device of any preceding example implementation, or any combination of any preceding example implementations, wherein the vibrating assembly further includes a piezoelectric component affixed to and substantially surrounding the mesh plate.

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The aerosol delivery device of any preceding example implementation, or any combination of any preceding example implementations, wherein the mesh plate is substantially flat.

The aerosol delivery device of any preceding example implementation, or any combination of any preceding example implementations, wherein at least a portion of the mesh plate is convex with respect to the reservoir.

An aerosol delivery device comprising a housing defining an outer wall, and further including a power source and a control component, a mouthpiece portion, a tank portion that includes a reservoir configured to contain a liquid composition, and an atomization assembly configured to vaporize the liquid composition to generate an aerosol, wherein the atomization assembly comprises a vibrating assembly that includes a mesh plate, and a pump assembly configured to transfer a portion of the liquid composition from the reservoir to the mesh plate.

The aerosol delivery device of any preceding example implementation, or any combination of any preceding example implementations, wherein the pump assembly is configured to deliver the liquid composition to the mesh plate on demand.

The aerosol delivery device of any preceding example implementation, or any combination of any preceding example implementations, wherein the pump assembly utilizes one or more nozzles to transfer the liquid composition to the mesh plate.

The aerosol delivery device of any preceding example implementation, or any combination of any preceding example implementations, wherein the pump assembly comprises at least one of a dispensing mechanism, a diaphragm device, and a peristaltic device.

The aerosol delivery device of any preceding example implementation, or any combination of any preceding example implementations, wherein the dispensing mechanism comprises a shape-memory mechanism.

The aerosol delivery device of any preceding example implementation, or any combination of any preceding example implementations, wherein the vibrating assembly further includes a piezoelectric component affixed to and substantially surrounding the mesh plate.

The aerosol delivery device of any preceding example implementation, or any combination of any preceding example implementations, wherein the mesh plate is substantially flat.

The aerosol delivery device of any preceding example implementation, or any combination of any preceding example implementations, wherein at least a portion of the mesh plate is convex with respect to the reservoir.

An aerosol delivery device comprising a housing defining an outer wall, and further including a power source and a control component, a mouthpiece portion, a tank portion that includes a reservoir configured to contain a liquid composition, and an atomization assembly configured to vaporize the liquid composition to generate an aerosol, wherein the atomization assembly comprises a vibrating assembly that includes a mesh plate, and a blower assembly configured to propel a portion of the liquid composition from the reservoir to the mesh plate.

The aerosol delivery device of any preceding example implementation, or any combination of any preceding example implementations, wherein the blower assembly comprises a compressor.

The aerosol delivery device of any preceding example implementation, or any combination of any preceding

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example implementations, wherein the blower assembly includes one or more nozzles.

The aerosol delivery device of any preceding example implementation, or any combination of any preceding example implementations, wherein the blower assembly utilizes a pressurized gas to propel the liquid composition to the mesh plate.

The aerosol delivery device of any preceding example implementation, or any combination of any preceding example implementations, wherein the pressurized gas comprises at least one of air, carbon dioxide (CO₂), and nitrogen (N₂).

The aerosol delivery device of any preceding example implementation, or any combination of any preceding example implementations, wherein the vibrating assembly further includes a piezoelectric component affixed to and substantially surrounding the mesh plate.

The aerosol delivery device of any preceding example implementation, or any combination of any preceding example implementations, wherein the mesh plate is substantially flat.

The aerosol delivery device of any preceding example implementation, or any combination of any preceding example implementations, wherein at least a portion of the mesh plate is convex with respect to the reservoir.

An aerosol delivery device comprising a housing defining an outer wall, and further including a power source and a control component, a mouthpiece portion, a tank portion that includes a reservoir configured to contain a liquid composition, and an atomization assembly configured to vaporize the liquid composition to generate an aerosol, wherein the atomization assembly comprises a vibrating assembly that includes a mesh plate, and the reservoir comprises a U-shaped tube comprising a first reservoir section, a second reservoir section, and a third reservoir section, wherein the third reservoir section is located proximate the atomization assembly, and wherein the second reservoir section connects the first reservoir section to the third reservoir section.

The aerosol delivery device of any preceding example implementation, or any combination of any preceding example implementations, wherein the first reservoir section has a larger diameter than the diameter of the second and third reservoir sections.

The aerosol delivery device of any preceding example implementation, or any combination of any preceding example implementations, further comprising a plummet configured to exert a downward force on the liquid composition in the first reservoir section.

The aerosol delivery device of any preceding example implementation, or any combination of any preceding example implementations, wherein the plummet comprises at least one of a weighted disk and a weighted ball.

The aerosol delivery device of any preceding example implementation, or any combination of any preceding example implementations, wherein the plummet further comprises an active component configured to exert a downward force on the plummet.

The aerosol delivery device of any preceding example implementation, or any combination of any preceding example implementations, wherein the reservoir is angled with respect to a longitudinal axis of the aerosol delivery device.

The aerosol delivery device of any preceding example implementation, or any combination of any preceding example implementations, wherein the mouthpiece portion

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is positioned on one side of the aerosol delivery device body so as to encourage use of the device in particular orientations.

The aerosol delivery device of any preceding example implementation, or any combination of any preceding example implementations, wherein the vibrating assembly further includes a piezoelectric component affixed to and substantially surrounding the mesh plate.

The aerosol delivery device of any preceding example implementation, or any combination of any preceding example implementations, wherein the mesh plate is substantially flat.

The aerosol delivery device of any preceding example implementation, or any combination of any preceding example implementations, wherein at least a portion of the mesh plate is convex with respect to the reservoir.

An aerosol delivery device comprising a housing defining an outer wall, and further including a power source and a control component, a mouthpiece portion, a tank portion that includes a reservoir configured to contain a liquid composition, and an atomization assembly configured to vaporize the liquid composition to generate an aerosol, wherein the atomization assembly comprises a vibrating assembly that includes a mesh plate, wherein the reservoir comprises a substantially cylindrical tube, and further comprising a plummet comprising an active component configured to exert a downward force on the liquid composition in the first reservoir section.

The aerosol delivery device of any preceding example implementation, or any combination of any preceding example implementations, wherein the active component comprises a spring.

These and other features, aspects, and advantages of the present disclosure will be apparent from a reading of the following detailed description together with the accompanying drawings, which are briefly described below. The present disclosure includes any combination of two, three, four or more features or elements set forth in this disclosure, regardless of whether such features or elements are expressly combined or otherwise recited in a specific example implementation described herein. This disclosure is intended to be read holistically such that any separable features or elements of the disclosure, in any of its aspects and example implementations, should be viewed as intended, namely to be combinable, unless the context of the disclosure clearly dictates otherwise.

It will therefore be appreciated that this Brief Summary is provided merely for purposes of summarizing some example implementations so as to provide a basic understanding of some aspects of the disclosure. Accordingly, it will be appreciated that the above described example implementations are merely examples and should not be construed to narrow the scope or spirit of the disclosure in any way. Other example implementations, aspects and advantages will become apparent from the following detailed description taken in conjunction with the accompanying drawings which illustrate, by way of example, the principles of some described example implementations.

BRIEF DESCRIPTION OF THE DRAWING(S)

In order to assist the understanding of aspects of the disclosure, reference will now be made to the appended drawings, which are not necessarily drawn to scale and in which like reference numerals refer to like elements. The drawings are provided by way of example to assist under-

standing of aspects of the disclosure, and should not be construed as limiting the disclosure.

FIG. 1 is a perspective schematic view of an aerosol delivery device comprising a cartridge and a control unit wherein the cartridge and control unit are shown in a coupled configuration, according to an example implementation of the present disclosure;

FIG. 2 illustrates a front cross-section schematic view of an aerosol delivery device comprising a cartridge and a control unit wherein the cartridge and control unit are shown in a de-coupled configuration, according to an example implementation of the present disclosure;

FIG. 3 illustrates a perspective view of a portion of an atomization assembly, according to an example implementation of the present disclosure;

FIG. 4A illustrates a side schematic view of a portion of an atomization assembly, according to an example implementation of the present disclosure;

FIG. 4B illustrates a side schematic view of a portion of an atomization assembly, according to an example implementation of the present disclosure;

FIG. 4C illustrates a side schematic view of a portion of an atomization assembly, according to an example implementation of the present disclosure;

FIG. 4D illustrates a side schematic view of a portion of an atomization assembly, according to an example implementation of the present disclosure;

FIG. 4E illustrates a side schematic view of a portion of an atomization assembly, according to an example implementation of the present disclosure;

FIG. 4F illustrates a side schematic view of a portion of an atomization assembly, according to an example implementation of the present disclosure;

FIG. 5 illustrates a side schematic view of a portion of a reservoir configured to hold a liquid composition, and an atomization assembly configured to generate an aerosol from the liquid composition, according to an example implementation of the present disclosure;

FIG. 6 illustrates a side schematic view of a portion of a reservoir configured to hold a liquid composition, and an atomization assembly configured to generate an aerosol from the liquid composition, according to an example implementation of the present disclosure;

FIG. 7 illustrates a perspective schematic view of a portion of a reservoir configured to hold a liquid composition, and an atomization assembly configured to generate an aerosol from the liquid composition, according to an example implementation of the present disclosure;

FIG. 8A illustrates a top schematic view of a portion of a reservoir assembly configured to hold a liquid composition, and an atomization assembly configured to generate an aerosol from the liquid composition, according to example implementations of the present disclosure;

FIG. 8B illustrates a top schematic view of a portion of a reservoir assembly configured to hold a liquid composition, and an atomization assembly configured to generate an aerosol from the liquid composition, according to example implementations of the present disclosure;

FIG. 8C illustrates a top schematic view of a portion of an aerosol delivery device including a reservoir assembly configured to hold a liquid composition, and an atomization assembly configured to generate an aerosol from the liquid composition, according to example implementations of the present disclosure;

FIG. 9 illustrates a perspective schematic view of various portions of an aerosol delivery device including a reservoir configured to hold a liquid composition, and an atomization

assembly configured to generate an aerosol from the liquid composition, according to an example implementation of the present disclosure;

FIG. 10 illustrates a top cross-section schematic view of a portion of a reservoir configured to hold a liquid composition, and an atomization assembly configured to generate an aerosol from the liquid composition, according to an example implementation of the present disclosure;

FIG. 11 illustrates a side schematic view of a portion of a reservoir configured to hold a liquid composition, and an atomization assembly configured to generate an aerosol from the liquid composition, according to an example implementation of the present disclosure;

FIG. 12 illustrates a side schematic view of a portion of a reservoir configured to hold a liquid composition, and an atomization assembly configured to generate an aerosol from the liquid composition, according to an example implementation of the present disclosure;

FIG. 13 illustrates a side schematic view of a portion of a reservoir configured to hold a liquid composition, and an atomization assembly configured to generate an aerosol from the liquid composition, according to an example implementation of the present disclosure;

FIG. 14 illustrates a side schematic view of a portion of a reservoir configured to hold a liquid composition, and an atomization assembly configured to generate an aerosol from the liquid composition, according to an example implementation of the present disclosure;

FIG. 15A illustrates a side schematic view of a portion of a reservoir configured to hold a liquid composition, and an atomization assembly configured to generate an aerosol from the liquid composition, according to an example implementation of the present disclosure;

FIG. 15B illustrates a side schematic view of a portion of a reservoir configured to hold a liquid composition, and an atomization assembly configured to generate an aerosol from the liquid composition, according to an example implementation of the present disclosure;

FIG. 16 illustrates a side schematic view of various portions of an aerosol delivery device including a reservoir configured to hold a liquid composition, and an atomization assembly configured to generate an aerosol from the liquid composition, according to an example implementation of the present disclosure; and

FIG. 17 illustrates a side schematic view of a portion of a reservoir configured to hold a liquid composition, and an atomization assembly configured to generate an aerosol from the liquid composition, according to an example implementation of the present disclosure.

DETAILED DESCRIPTION

The present disclosure will now be described more fully hereinafter with reference to example embodiments thereof. These example embodiments are described so that this disclosure will be thorough and complete, and will fully convey the scope of the disclosure to those skilled in the art. Indeed, the disclosure may be embodied in many different forms and should not be construed as limited to the embodiments set forth herein; rather, these embodiments are provided so that this disclosure will satisfy applicable legal requirements. As used in the specification, and in the appended claims, the singular forms “a”, “an”, “the”, include plural referents unless the context clearly dictates otherwise.

As described hereinafter, embodiments of the present disclosure relate to aerosol delivery devices or vaporization

devices, said terms being used herein interchangeably. Aerosol delivery devices according to the present disclosure use electrical energy to vaporize a material (preferably without combusting the material to any significant degree and/or without significant chemical alteration of the material) to form an inhalable substance; and components of such devices have the form of articles that most preferably are sufficiently compact to be considered hand-held devices. That is, use of components of some aerosol delivery devices does not result in the production of smoke—i.e., from by-products of combustion or pyrolysis of tobacco, but rather, use of those preferred systems results in the production of vapors resulting from vaporization of an aerosol precursor composition. In some examples, components of aerosol delivery devices may be characterized as electronic cigarettes, and those electronic cigarettes most preferably incorporate tobacco and/or components derived from tobacco, and hence deliver tobacco derived components in aerosol form. Other examples include delivery devices for Tetrahydrocannabinol (THC), Cannabidiol (CBD), botanicals, medicinals, and/or other active ingredients.

Aerosol generating devices of certain preferred aerosol delivery devices may provide many of the sensations (e.g., inhalation and exhalation rituals, types of tastes or flavors, organoleptic effects, physical feel, use rituals, visual cues such as those provided by visible aerosol, and the like) of smoking a cigarette, cigar, or pipe that is employed by lighting and burning tobacco (and hence inhaling tobacco smoke), without any substantial degree of combustion of any component thereof. For example, the user of an aerosol generating device of the present disclosure can hold and use the device much like a smoker employs a traditional type of smoking article, draw on one end of that device for inhalation of aerosol produced by that device, take or draw puffs at selected intervals of time, and the like.

Aerosol delivery devices of the present disclosure also may be characterized as being vapor-producing articles or medicament delivery articles. Thus, such articles or devices may be adapted so as to provide one or more substances (e.g., flavors and/or pharmaceutical active ingredients) in an inhalable form or state. For example, inhalable substances may be substantially in the form of a vapor (i.e., a substance that is in the gas phase at a temperature lower than its critical point). Alternatively, inhalable substances may be in the form of an aerosol (i.e., a suspension of fine solid particles or liquid droplets in a gas). For purposes of simplicity, the term “aerosol” as used herein is meant to include vapors, gases, and aerosols of a form or type suitable for human inhalation, whether or not visible, and whether or not of a form that might be considered to be smoke-like.

Aerosol delivery devices of the present disclosure most preferably comprise some combination of a power source (i.e., an electrical power source), at least one control component (e.g., means for actuating, controlling, regulating and ceasing power, such as by controlling electrical current flow the power source to other components of the article—e.g., a microcontroller or microprocessor), an atomization assembly, a liquid composition (e.g., commonly an aerosol precursor composition liquid capable of yielding an aerosol, such as ingredients commonly referred to as “smoke juice,” “e-liquid” and “e-juice”), and a mouthpiece or mouth region for allowing draw upon the aerosol delivery device for aerosol inhalation (e.g., a defined airflow path through the article such that aerosol generated may be withdrawn therefrom upon draw).

Alignment of the components within the aerosol delivery device may be variable. In specific embodiments, the aerosol

precursor composition may be located between two opposing ends of the device (e.g., within a reservoir of a cartridge, which in certain circumstances is replaceable and disposable or refillable). Other configurations, however, are not excluded. Generally, the components are configured relative to one another so that energy from the atomization assembly vaporizes the aerosol precursor composition (as well as one or more flavorants, medicaments, or the like that may likewise be provided for delivery to a user) and forms an aerosol for delivery to the user. When the atomization assembly vaporizes the aerosol precursor composition, an aerosol is formed, released, or generated in a physical form suitable for inhalation by a consumer. It should be noted that the foregoing terms are meant to be interchangeable such that reference to release, releasing, releases, or released includes form or generate, forming or generating, forms or generates, and formed or generated. Specifically, an inhalable substance is released in the form of a vapor or aerosol or mixture thereof.

More specific formats, configurations and arrangements of components within the aerosol delivery devices of the present disclosure will be evident in light of the further disclosure provided hereinafter. Additionally, the selection and arrangement of various aerosol delivery device components may be appreciated upon consideration of the commercially available electronic aerosol delivery devices, such as those representative products referenced in the background art section of the present disclosure.

FIG. 1 illustrates an aerosol delivery device comprising a cartridge and a control unit wherein the cartridge and control unit are shown in a coupled configuration, according to an example implementation of the present disclosure. In particular, FIG. 1 illustrates a perspective schematic view of an aerosol delivery device 100 comprising a cartridge 104 and a control unit 102. As depicted in the figure, the cartridge 104 may be permanently or detachably aligned in a functioning relationship with the control unit 102. In some implementations, for example, the cartridge and the control unit may comprise a single part, whereas in other implementations (such as the depicted implementation), a connection therebetween may be releasable such that, for example, the control unit may be reused with one or more additional cartridges that may be disposable and/or refillable. In various implementations, a variety of different means of engagement may be used to couple a cartridge and a control unit together. For example, in some implementations the cartridge and the control unit may be coupled via one or more of a snap fit engagement, a press fit engagement, a threaded engagement, and a magnetic engagement. It should be noted that the components depicted in this and the other figures are representative of the components that may be present in a control unit and/or cartridge and are not intended to limit the scope of the control unit and/or cartridge components that are encompassed by the present disclosure.

FIG. 2 illustrates a front cross-section schematic view of the aerosol delivery device 100, wherein the cartridge 104 and control unit 102 of FIG. 1 are shown in a de-coupled configuration. In various implementations, the aerosol delivery device 100 may have a variety of different shapes. For example, in some implementations (such as the depicted implementation) the aerosol delivery device 100 may be substantially rod-like or substantially tubular shaped or substantially cylindrically shaped. In other implementations, however, other shapes and dimensions are possible (e.g., rectangular, oval, hexagonal, prismatic, regular or irregular polygon shapes, disc-shaped, cube-shaped, multifaceted

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shapes, or the like). In still other implementations, the cartridge and the control unit may have different shapes.

In the depicted implementation, the control unit **102** and the cartridge **104** include components adapted to facilitate mechanical engagement therebetween. Although a variety of other configurations are possible, the control unit **102** of the depicted implementation includes a coupler **124** that defines a cavity **125** therein. Likewise, the cartridge **104** includes a base **140** adapted to engage the coupler **124** of the control unit **102**. A coupler and a base that may be useful according to the present disclosure are described in U.S. Pat. App. Pub. No. 2014/0261495 to Novak et al., the disclosure of which is incorporated herein by reference in its entirety.

It should be noted, however, that in other implementations various other structures, shapes, and/or components may be employed to couple the control unit and the cartridge. For example, in some implementations the control unit and cartridge may be coupled together via an interference or press fit connection such as, for example, implementations wherein the control body includes a chamber configured to receive at least a portion of the cartridge or implementations wherein the cartridge includes a chamber configured to receive at least a portion of the control unit. In other implementations, the cartridge and the control unit may be coupled together via a screw thread connection. In still other implementations, the cartridge and the control unit may be coupled together via a bayonet connection. In still other implementations, the cartridge and the control unit may be coupled via a magnetic connection. In various implementations, once coupled an electrical connection may be created between the cartridge and the control unit so as to electrically connect the cartridge (and components thereof) to the battery and/or via the control component of the control unit. Such an electrical connection may exist via one or more components of the coupling features. In such a manner, corresponding electrical contacts in the cartridge and the control unit may be substantially aligned after coupling to provide the electrical connection.

In specific implementations, one or both of the control unit **102** and the cartridge **104** may be referred to as being disposable or as being reusable. For example, in some implementations the control unit may have a replaceable battery or a rechargeable battery and thus may be combined with any type of recharging technology, including connection to a wall charger, connection to a car charger (e.g., cigarette lighter receptacle, USB port, etc.), connection to a computer, any of which may include a universal serial bus (USB) cable or connector (e.g., USB 2.0, 3.0, 3.1, USB Type-C), connection to a USB connector (e.g., USB 2.0, 3.0, 3.1, USB Type-C as may be implemented in a wall outlet, electronic device, vehicle, etc.), connection to a photovoltaic cell (sometimes referred to as a solar cell) or solar panel of solar cells, or wireless charger, such as a charger that uses inductive wireless charging (including for example, wireless charging according to the Qi wireless charging standard from the Wireless Power Consortium (WPC)), or a wireless radio frequency (RF) based charger, and connection to an array of external cell(s) such as a power bank to charge a device via a USB connector or a wireless charger. An example of an inductive wireless charging system is described in U.S. Pat. App. Pub. No. 2017/0112196 to Sur et al., which is incorporated herein by reference in its entirety. In further implementations, a power source may also comprise a capacitor. Capacitors are capable of discharging more quickly than batteries and can be charged between puffs, allowing the battery to discharge into the capacitor at a lower rate. For example, a supercapacitor—

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e.g., an electric double-layer capacitor (EDLC)—may be used separate from or in combination with a battery. When used alone, the supercapacitor may be recharged before each use of the article. Thus, the device may also include a charger component that can be attached to the smoking article between uses to replenish the supercapacitor. Examples of power supplies that include supercapacitors are described in U.S. Pat. App. Pub. No. 2017/0112191 to Sur et al., which is incorporated herein by reference in its entirety.

As illustrated in the figure, the control unit **102** may be formed of a control unit housing **101** that includes a control component **106** (e.g., a printed circuit board (PCB), an integrated circuit, a memory component, a microcontroller, or the like), a flow sensor **108**, a battery **110**, and a light-emitting diode (LED) **112**, which components may be variably aligned. Some example types of electronic components, structures, and configurations thereof, features thereof, and general methods of operation thereof, are described in U.S. Pat. No. 4,735,217 to Gerth et al.; U.S. Pat. No. 4,947,874 to Brooks et al.; U.S. Pat. No. 5,372,148 to McCafferty et al.; U.S. Pat. No. 6,040,560 to Fleischhauer et al.; U.S. Pat. No. 7,040,314 to Nguyen et al. and U.S. Pat. No. 8,205,622 to Pan; U.S. Pat. App. Pub. Nos. 2009/0230117 to Fernando et al., 2014/0060554 to Collet et al., and 2014/0270727 to Ampolini et al.; and U.S. Pat. App. Pub. No. 2015/0257445 to Henry et al.; which are incorporated herein by reference in their entireties. Some examples of batteries that may be applicable to the present disclosure are described in U.S. Pat. App. Pub. No. 2010/0028766 to Peckerar et al., the disclosure of which is incorporated herein by reference in its entirety. In some implementations, further indicators (e.g., a haptic feedback component, an audio feedback component, or the like) may be included in addition to or as an alternative to the LED. Additional representative types of components that yield visual cues or indicators, such as light emitting diode (LED) components, and the configurations and uses thereof, are described in U.S. Pat. No. 5,154,192 to Sprinkel et al.; U.S. Pat. No. 8,499,766 to Newton and U.S. Pat. No. 8,539,959 to Scatterday; U.S. Pat. App. Pub. No. 2015/0020825 to Galloway et al.; and U.S. Pat. App. Pub. No. 2015/0216233 to Sears et al.; which are incorporated herein by reference in their entireties. It should be understood that in various implementations not all of the illustrated elements may be required. For example, in some implementations an LED may be absent or may be replaced with a different indicator, such as a vibrating indicator. Likewise, a flow sensor may be replaced with a manual actuator, such as, for example, one or more manually actuated push buttons.

In the depicted implementation, the cartridge **104** may be formed of a cartridge housing **103**, which may define a liquid reservoir **144** configured to contain a liquid composition **145**. In some implementations, the liquid reservoir may be part of the cartridge housing (such as, for example, comprising a molded feature of the cartridge housing), while in other implementations, the liquid reservoir may comprise a separate part. In some implementations, the liquid reservoir may be disposable. In other implementations, the liquid reservoir may be refillable. In various implementations, the liquid composition contained in the liquid reservoir **144** may comprise an aerosol precursor composition. Some examples of types of substrates, reservoirs, or other components for supporting a liquid composition are described in U.S. Pat. No. 8,528,569 to Newton; U.S. Pat. App. Pub. Nos. 2014/0261487 to Chapman et al. and 2014/0059780 to Davis et al.; and U.S. Pat. App. Pub. No. 2015/0216232 to Bless et

al.; which are incorporated herein by reference in their entireties. Additionally, various wicking materials, and the configuration and operation of those wicking materials within certain types of electronic cigarettes, are set forth in U.S. Pat. No. 8,910,640 to Sears et al.; which is incorporated herein by reference in its entirety.

In some implementations, the reservoir may be made of a polymeric material that, in further implementations, may be at least partially transparent or translucent. In some implementations, such materials, may include, but need not be limited to, polycarbonate, acrylic, polyethylene terephthalate (PET), amorphous copolyester (PETG), polyvinyl chloride (PVC), liquid silicone rubber (LSR), cyclic olefin copolymers, polyethylene (PE), ionomer resin, polypropylene (PP), fluorinated ethylene propylene (FEP), styrene methyl methacrylate (SMMA), styrene acrylonitrile resin (SAN), polystyrene, acrylonitrile butadiene styrene (ABS), and combinations thereof. In other implementations, the reservoir may be made of other material that may be at least partially transparent or translucent. Such materials may include, for example, glass or ceramic materials.

In some implementations, the aerosol precursor composition may incorporate nicotine, which may be present in various concentrations. The source of nicotine may vary, and the nicotine incorporated in the aerosol precursor composition may derive from a single source or a combination of two or more sources. For example, in some implementations the aerosol precursor composition may include nicotine derived from tobacco. In other implementations, the aerosol precursor composition may include nicotine derived from other organic plant sources, such as, for example, non-tobacco plant sources including plants in the Solanaceae family. In other implementations, the aerosol precursor composition may include synthetic nicotine. In some implementations, nicotine incorporated in the aerosol precursor composition may be derived from non-tobacco plant sources, such as other members of the Solanaceae family. In some implementations, the aerosol precursor composition may additionally or alternatively include alcohol, other botanical substances, other medicinal substances, or may include Tetrahydrocannabinol (THC), Cannabidiol (CBD), or other active ingredients, or some combination thereof.

In some implementations, the aerosol precursor composition may incorporate tobacco or components derived from tobacco. In one regard, the tobacco may be provided as parts or pieces of tobacco, such as finely ground, milled or powdered tobacco lamina. Tobacco beads, pellets, or other solid forms may be included, such as described in U.S. Pat. App. Pub. No. 2015/0335070 to Sears et al., the disclosure of which is incorporated herein by reference in its entirety. In another regard, the tobacco may be provided in the form of an extract, such as a spray dried extract that incorporates many of the water soluble components of tobacco. Alternatively, tobacco extracts may have the form of relatively high nicotine content extracts, which extracts also incorporate minor amounts of other extracted components derived from tobacco. In another regard, components derived from tobacco may be provided in a relatively pure form, such as certain flavoring agents that are derived from tobacco. In one regard, a component that is derived from tobacco, and that may be employed in a highly purified or essentially pure form, is nicotine (e.g., pharmaceutical grade nicotine, USP/EP nicotine, etc.). In other implementations, non-tobacco materials alone may form the aerosol precursor composition. In some implementations, the aerosol precursor composition may include tobacco-extracted nicotine with tobacco or

non-tobacco flavors and/or non-tobacco-extracted nicotine with tobacco or non-tobacco flavors.

In the depicted implementation, the liquid composition, sometimes referred to as an aerosol precursor composition or a vapor precursor composition or “e-liquid”, may comprise a variety of components, which may include, by way of example, a polyhydric alcohol (e.g., glycerin, propylene glycol, or a mixture thereof), nicotine, tobacco, tobacco extract, and/or flavorants. Representative types of aerosol precursor components and formulations are also set forth and characterized in U.S. Pat. No. 7,217,320 to Robinson et al. and U.S. Pat. App. Pub. Nos. 2013/0008457 to Zheng et al.; 2013/0213417 to Chong et al.; 2014/0060554 to Collett et al.; 2015/0020823 to Lipowicz et al.; and 2015/0020830 to Koller, as well as WO 2014/182736 to Bowen et al, the disclosures of which are incorporated herein by reference in their entireties. Other aerosol precursors that may be employed include the aerosol precursors that have been incorporated in VUSE® products by R. J. Reynolds Vapor Company, the BLU™ products by Fontem Ventures B.V., the MISTIC MENTHOL product by Mistic Ecigs, MARK TEN products by Nu Mark LLC, the JUUL product by Juul Labs, Inc., and VYPE products by CN Creative Ltd. Also desirable are the so-called “smoke juices” for electronic cigarettes that have been available from Johnson Creek Enterprises LLC. Still further example aerosol precursor compositions are sold under the brand names BLACK NOTE, COSMIC FOG, THE MILKMAN E-LIQUID, FIVE PAWNS, THE VAPOR CHEF, VAPE WILD, BOOSTED, THE STEAM FACTORY, MECH SAUCE, CASEY JONES MAINLINE RESERVE, MITTEN VAPORS, DR. CRIMMY’S V-LIQUID, SMILEY E LIQUID, BEANTOWN VAPOR, CUTTWOOD, CYCLOPS VAPOR, SICBOY, GOOD LIFE VAPOR, TELEOS, PINUP VAPORS, SPACE JAM, MT. BAKER VAPOR, and JIMMY THE JUICE MAN.

The amount of aerosol precursor that is incorporated within the aerosol delivery system is such that the aerosol generating device provides acceptable sensory and desirable performance characteristics. For example, it is highly preferred that sufficient amounts of aerosol forming material (e.g., glycerin and/or propylene glycol), be employed in order to provide for the generation of a visible mainstream aerosol that in many regards resembles the appearance of tobacco smoke. The amount of aerosol precursor within the aerosol generating system may be dependent upon factors such as the number of puffs desired per aerosol generating device. In one or more embodiments, about 1 ml or more, about 2 ml or more, about 5 ml or more, or about 10 ml or more of the aerosol precursor composition may be included.

In the some of the examples described above, the aerosol precursor composition comprises a glycerol-based liquid. In other implementations, however, the aerosol precursor composition may be a water-based liquid. In some implementations, the water-based liquid may be comprised of more than approximately 80% water. For example, in some implementations the percentage of water in the water-based liquid may be in the inclusive range of approximately 90% to approximately 93%. In some implementations, the water-based liquid may include up to approximately 10% propylene glycol. For example, in some implementations the percentage of propylene glycol in the water-based liquid may be in the inclusive range of approximately 4% to approximately 5%. In some implementations, the water-based liquid may include up to approximately 10% flavorant. For example, in some implementations the percentage of flavorant(s) of the water-based liquid may be in the

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inclusive range of approximately 3% to approximately 7%. In some implementations, the water-based liquid may include up to approximately 1% nicotine. For example, in some implementations the percentage nicotine in the water-based liquid may be in the inclusive range of approximately 0.1% to approximately 0.3%. In some implementations, the water-based liquid may include up to approximately 10% cyclodextrin. For example, in some implementations the percentage cyclodextrin in the water-based liquid may be in the inclusive range of approximately 3% to 5%. In still other implementations, the aerosol precursor composition may be a combination of a glycerol-based liquid and a water-based liquid. For example, some implementations may include up to approximately 50% water and less than approximately 20% glycerol. The remaining components may include one or more of propylene glycol, flavorants, nicotine, cyclodextrin, etc. Some examples of water-based liquid compositions that may be suitable are disclosed in GB 1817863.2, filed Nov. 1, 2018, titled Aerosolizable Formulation; GB 1817864.0, filed Nov. 1, 2018, titled Aerosolizable Formulation; GB 1817867.3, filed Nov. 1, 2018, titled Aerosolizable Formulation; GB 1817865.7, filed Nov. 1, 2018, titled Aerosolizable Formulation; GB 1817859.0, filed Nov. 1, 2018, titled Aerosolizable Formulation; GB 1817866.5, filed Nov. 1, 2018, titled Aerosolizable Formulation; GB 1817861.6, filed Nov. 1, 2018, titled Gel and Crystalline Powder; GB 1817862.4, filed Nov. 1, 2018, titled Aerosolizable Formulation; GB 1817868.1, filed Nov. 1, 2018, titled Aerosolized Formulation; and GB 1817860.8, filed Nov. 1, 2018, titled Aerosolized Formulation, each of which is incorporated by reference herein in its entirety.

As noted above, in various implementations the liquid composition may include a flavorant. In some implementations, the flavorant may be pre-mixed with the liquid. In other implementations, the flavorant may be delivered separately downstream from the atomizer as a main or secondary flavor. Still other implementations may combine a pre-mixed flavorant with a downstream flavorant. As used herein, reference to a "flavorant" refers to compounds or components that can be aerosolized and delivered to a user and which impart a sensory experience in terms of taste and/or aroma. Example flavorants include, but are not limited to, vanillin, ethyl vanillin, cream, tea, coffee, fruit (e.g., apple, cherry, strawberry, peach and citrus flavors, including lime, lemon, mango, and other citrus flavors), maple, menthol, mint, peppermint, spearmint, wintergreen, nutmeg, clove, lavender, cardamom, ginger, honey, anise, sage, rosemary, hibiscus, rose hip, yerba mate, guayusa, honeybush, rooibos, amaretto, mojito, yerba santa, ginseng, chamomile, turmeric, bacopa monniera, ginkgo biloba, withania somnifera, cinnamon, sandalwood, jasmine, cascarilla, cocoa, licorice, and flavorings and flavor packages of the type and character traditionally used for the flavoring of cigarette, cigar, and pipe tobaccos. Other examples include flavorants derived from, or simulating, burley, oriental tobacco, flue cured tobacco, etc. Syrups, such as high fructose corn syrup, also can be employed. Example plant-derived compositions that may be suitable are disclosed in U.S. Pat. No. 9,107,453 and U.S. Pat. App. Pub. No. 2012/0152265 both to Dube et al., the disclosures of which are incorporated herein by reference in their entireties. The selection of such further components are variable based upon factors such as the sensory characteristics that are desired for the smoking article, and the present disclosure is intended to encompass any such further components that are readily apparent to those skilled in the art of tobacco and tobacco-related or tobacco-derived products. See, e.g., Gutcho, Tobacco Fla-

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voring Substances and Methods, Noyes Data Corp. (1972) and Leffingwell et al., Tobacco Flavoring for Smoking Products (1972), the disclosures of which are incorporated herein by reference in their entireties. It should be noted that reference to a flavorant should not be limited to any single flavorant as described above, and may, in fact, represent a combination of one or more flavorants.

Referring back to FIG. 2, the liquid reservoir **144** of the depicted implementation may be in fluid communication with (either directly or through one or more additional components) at least a portion of an atomization assembly **115**. As will be discussed in more detail below, in some implementations, the liquid reservoir **144** may comprise an independent container (e.g., formed of walls substantially impermeable to the liquid composition), which, in some implementations, may be configured to rotate relative to the body of the aerosol delivery device (e.g., the control housing unit and/or the cartridge housing unit). In some implementations, the walls of the liquid reservoir may be flexible and/or collapsible, while in other implementations the walls of the liquid reservoir may be substantially rigid. In some implementations, the liquid reservoir may be substantially sealed to prevent passage of the liquid composition therefrom except via any specific openings or conduits provided expressly for passage of the liquid composition, such as through one or more transport elements as otherwise described herein. An electrical connection **116** connects the atomization assembly **115** to control component **106** and/or the battery **110**. In the depicted implementation, the atomization assembly **115** is connected to the base **140** of the cartridge **104**, which, when assembled to the control unit **102**, provides an electrical connection to the control component **106** and/or the battery **110**. As noted, the atomization assembly **115** is configured to be electrically connected to the battery **110** and/or the control component **106**. In such a manner, the atomization assembly **115** of the depicted implementation may be energized by the battery **110** and/or control component **106** (e.g., so as to vibrate a component of the atomization assembly at a relatively high rate). In various implementations, an atomization assembly **115** may be fluidly coupled with a portion of the liquid composition such that the atomization assembly **115** generates an aerosol from the contacted liquid composition. In various implementations, an atomization assembly may be directly fluidly coupled with a portion of the liquid composition, or indirectly fluidly coupled with a portion of the liquid composition, such as via a liquid transport element. In various implementations, a liquid transport element may have one layer, or multiple layers, and may be made of a single material or multiple materials. In various implementations, the liquid transport element may be any shape and may be a porous, semi-porous, or non-porous absorbent/adsorbent material.

For example, in some implementations the liquid transport element may be made of fibrous materials (e.g., organic cotton, cellulose acetate, regenerated cellulose fabrics, glass fibers), polymers, silk, particles, porous ceramics (e.g., alumina, silica, zirconia, SiC, SiN, AlN, etc.), porous metals, porous carbon, graphite, porous glass, sintered glass beads, sintered ceramic beads, capillary tubes, porous polymers, or the like. In some implementations, the liquid transport element may be any material that contains an open pore network (i.e., a plurality of pores that are interconnected so that fluid may flow from one pore to another in a plurality of direction through the element). The pores can be nanopores, micropores, macropores or combinations thereof. As further discussed herein, some implementations of the pres-

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ent disclosure may particularly relate to the use of non-fibrous transport elements. As such, fibrous transport elements may be expressly excluded. Alternatively, combinations of fibrous transport elements and non-fibrous transport elements may be utilized. In some embodiments, the liquid transport element may be a substantially solid non-porous material, such as a polymer or dense ceramic or metals, configured to channel liquid through apertures or slots while not necessarily relying upon wicking through capillary action. Such a solid body may be used in combination with a porous absorptive pad. The absorptive pad may be formed of silica-based fibers, organic cotton, rayon fibers, cellulose acetate, regenerated cellulose fabrics, highly porous ceramic or metal mesh, etc. Some representative types of substrates, reservoirs or other components for supporting the aerosol precursor are described in U.S. Pat. No. 8,528,569 to Newton; U.S. Pat. App. Pub. Nos. 2014/0261487 to Chapman et al. and 2014/0059780 to Davis et al.; and U.S. Pat. App. Pub. No. 2015/0216232 to Bless et al.; which are incorporated herein by reference in their entireties. Additionally, various wicking materials, and the configuration and operation of those wicking materials within certain types of electronic cigarettes, are set forth in U.S. Pat. No. 8,910,640 to Sears et al.; which is incorporated herein by reference in its entirety. In some implementations, the liquid transport element may be formed partially or completely from a porous monolith, such as a porous ceramic, a porous glass, or the like. Example monolithic materials that may be suitable for use according to embodiments of the present disclosure are described, for example, in U.S. Pat. App. Pub. No. 2017/0188626 to Davis et al., and U.S. Pat. App. Pub. No. 2014/0123989 to LaMothe, the disclosures of which are incorporated herein by reference in their entireties. In some implementations, the porous monolith may form a substantially solid wick.

In various implementations, an end of the liquid transport element may be configured to be placed proximate the mesh plate and between the mesh plate and liquid composition in the reservoir so that the liquid transport element acts as a secondary reservoir that absorbs or adsorbs the liquid from the reservoir so that the mesh plate is in contact with the liquid composition, even if there is no longer liquid in the reservoir. In such a manner, the liquid transport element is configured to facilitate contact between the liquid composition and the atomization assembly.

In some implementations, the liquid composition may be driven through a component of the atomization assembly resulting in the generation of a plurality of aerosol particles. Likewise, in other implementations, vibration of a component of the atomization assembly may create ultrasonic waves within the liquid composition and/or surface acoustic waves in the liquid composition, that result in the formation of an aerosol at the surface of the liquid composition. As will be described in more detail below, in some implementations the liquid composition may be applied and/or transferred to a component of the atomization assembly to create the aerosol.

In the depicted implementation, the control unit housing **101** includes an air intake **118**, which may comprise an opening in the housing proximate the coupler **124** allowing for passage of ambient air into the control unit housing **101** where it then passes through the cavity **125** of the coupler **124**, and eventually into or around the atomization assembly **115**, where it may be mixed with the vaporized aerosol precursor composition to comprise the aerosol that is delivered to the user. It should be noted that in other implementations the air intake **118** is not limited being on or adjacent

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the control unit housing **101**. For example, in some implementations, an air intake may be formed through the cartridge housing **103** (e.g., such that it does not enter the control unit **102**) or some other portion of the aerosol delivery device **100**. In the depicted implementation, a mouthpiece portion that includes an opening **128** may be present in the cartridge housing **103** (e.g., at a mouthend of the cartridge **104**) to allow for egress of the formed aerosol from the cartridge **104**, such as for delivery to a user drawing on the mouthend of the cartridge **104**.

In various implementations, the cartridge **104** may also include at least one electronic component **150**, which may include an integrated circuit, a memory component, a sensor, or the like, although such a component need not be included. In those implementations that include such a component, the electronic component **150** may be adapted to communicate with the control component **106** and/or with an external device by wired or wireless means. In various implementations, the electronic component **150** may be positioned anywhere within the cartridge **104** or its base **140**. Some examples of electronic/control components that may be applicable to the present disclosure are described in U.S. Pat. App. Pub. No. 2019/0014819 to Sur, which is incorporated herein by reference in its entirety. Although in the depicted implementation the control component **106** and the flow sensor **108** are illustrated separately, it should be noted that in some implementations the control component and the flow sensor may be combined as an electronic circuit board with the air flow sensor attached directly thereto. In some embodiments, the air flow sensor may comprise its own circuit board or other base element to which it can be attached. In some embodiments, a flexible circuit board may be utilized. A flexible circuit board may be configured into a variety of shapes, include substantially tubular shapes. Configurations of a printed circuit board and a pressure sensor, for example, are described in U.S. Pat. App. Pub. No. 2015/0245658 to Worm et al., the disclosure of which is incorporated herein by reference. Additional types of sensing or detection mechanisms, structures, and configuration thereof, components thereof, and general methods of operation thereof, are described in U.S. Pat. No. 5,261,424 to Sprinkel, Jr.; U.S. Pat. No. 5,372,148 to McCafferty et al.; and PCT WO 2010/003480 to Flick; which are incorporated herein by reference in their entireties.

In some implementations, when a user draws on the article **100**, airflow may be detected by the sensor **108**, and the atomization assembly **115** may be activated, which may vaporize the liquid composition. As noted above, in some implementations drawing upon the mouthend of the article **100** causes ambient air to enter the air intake **118** and pass through the cavity **125** in the coupler **124** and the base **140**. In the cartridge **104**, the drawn air combines with the formed vapor to form the aerosol. The aerosol is whisked, aspirated, or otherwise drawn away from the atomization assembly **115** and out of the mouth opening **128** in the mouthend of the article **100**. As noted, in other implementations, in the absence of an airflow sensor, the atomization assembly **115** may be activated manually, such as by a push button (not shown). Additionally, in some implementations, the air intake may occur through the cartridge or between the cartridge and the control unit. It should be noted that in some implementations, there may be one or more components between the atomization assembly and the opening in the mouthend of the article. For example, in the depicted implementation a heating component **147** is located downstream from the atomization assembly **115**. In various implementations, the heating component may comprise any

device configured to elevate the temperature of the generated aerosol, including, for example, one or more coil heating components, ceramic heating components, etc.

In some implementations, one or more input elements may be included with the aerosol delivery device (and may replace or supplement an airflow sensor, pressure sensor, or manual push button). In various implementations, an input element may be included to allow a user to control functions of the device and/or for output of information to a user. Any component or combination of components may be utilized as an input for controlling the function of the device. For example, one or more pushbuttons may be used as described in U.S. Pat. App. Pub. No. 2015/0245658 to Worm et al., which is incorporated herein by reference in its entirety. Likewise, a touchscreen may be used as described in U.S. patent application Ser. No. 14/643,626, filed Mar. 10, 2015, to Sears et al., which is incorporated herein by reference in its entirety. As a further example, components adapted for gesture recognition based on specified movements of the aerosol delivery device may be used as an input. See U.S. App. Pub. No. 2016/0158782 to Henry et al., which is incorporated herein by reference in its entirety. As still a further example, a capacitive sensor may be implemented on the aerosol delivery device to enable a user to provide input, such as by touching a surface of the device on which the capacitive sensor is implemented.

In some embodiments, an input element may comprise a computer or computing device, such as a smartphone or tablet. In particular, the aerosol delivery device may be wired to the computer or other device, such as via use of a USB cord or similar protocol. The aerosol delivery device also may communicate with a computer or other device acting as an input via wireless communication. See, for example, the systems and methods for controlling a device via a read request as described in U.S. Pat. App. Pub. No. 2016/0007561 to Ampolini et al., the disclosure of which is incorporated herein by reference in its entirety. In such implementations, an APP or other computer program may be used in connection with a computer or other computing device to input control instructions to the aerosol delivery device, such control instructions including, for example, the ability to form an aerosol of specific composition by choosing the nicotine content and/or content of further flavors to be included.

Yet other features, controls or components that may be incorporated into aerosol delivery systems of the present disclosure are described in U.S. Pat. No. 5,967,148 to Harris et al.; U.S. Pat. No. 5,934,289 to Watkins et al.; U.S. Pat. No. 5,954,979 to Counts et al.; U.S. Pat. No. 6,040,560 to Fleischhauer et al.; U.S. Pat. No. 8,365,742 to Hon; U.S. Pat. No. 8,402,976 to Fernando et al.; U.S. Pat. App. Pub. Nos. 2010/0163063 to Fernando et al.; 2013/0192623 to Tucker et al.; 2013/0298905 to Leven et al.; 2013/0180553 to Kim et al.; 2014/0000638 to Sebastian et al.; 2014/0261495 to Novak et al.; and 2014/0261408 to DePiano et al.; which are incorporated herein by reference in their entireties.

In various implementations, the atomization assembly may comprise a variety of different components or devices configured to generate an aerosol from the liquid composition. For example, in some implementations the atomization assembly may comprise a jet nebulizer assembly, which may be configured to utilize compressed air to generate an aerosol. In other implementations, the atomization assembly may comprise an ultrasonic assembly, which may be configured to utilize the formation of ultrasonic waves within the liquid composition to generate an aerosol. In other implementations, the atomization assembly may comprise a

vibrating mesh assembly, which may comprise a piezoelectric material (e.g., a piezoelectric ceramic material), which may be affixed to and/or substantially surrounding a mesh plate, (e.g., a perforated plate) that is vibrated within the liquid composition or proximate the surface of the liquid composition to generate an aerosol. In other implementations, the atomization assembly may comprise a surface acoustic wave (SAW) or Raleigh wave assembly, which may utilize surface wave characteristics to generate an aerosol at the surface of the liquid composition. It should be noted that for purpose of this application, an ultrasonic assembly may be any assembly configured to create ultrasonic waves within the liquid composition. In some implementations, for example, a vibrating mesh assembly may also operate as an ultrasonic assembly.

An example of an atomization assembly of one implementation is shown in FIG. 3. In particular, FIG. 3 illustrates an atomization assembly **215** that comprises a piezoelectric ring **217** affixed to and substantially surrounding a mesh plate **219**. In some implementations, additional components may be included. For example, in some implementations a supporting component may be included that is located on the side of the mesh plate opposite the vibrating component (e.g., such that the mesh plate is sandwiched between the supporting component and the vibrating component). Although other configurations are possible, in some implementations, the supporting component may comprise a supporting ring. In various implementations, the supporting component may be made of any suitable material, including, but not limited to, polymeric, metal, and/or ceramic materials. In such a manner, in some implementations the supporting component may increase the longevity of the mesh plate. In some implementations, the supporting component may be replaceable, while in other implementations the supporting component may be affixed to the mesh plate and/or the vibrating component. In some implementations, an auxiliary component may be used that is located between mesh plate and the vibrating component. Although other configurations are possible, in some implementations, the auxiliary component may comprise an auxiliary ring. In various implementations, the auxiliary component may be made of any suitable material, including, but not limited to, polymeric, metal, and/or ceramic materials. In such a manner, the auxiliary component may facilitate the interfacial contact of the components. In some implementations, the auxiliary component may be replaceable, while in other implementations the auxiliary component may be affixed to the mesh plate and/or the vibrating component.

In various implementations, the vibrating component and the mesh plate may be permanently affixed to each other such as, for example, by affixing the components together via an adhesive, such as, for example, an epoxy or other glue, or by ultrasonic welding, mechanical fasteners, etc. It should be noted that while the depicted implementation describes a vibrating component in the form of a piezoelectric ring, in other implementations the vibrating component need not be limited to a ring-shaped object. For example, in some implementations the piezoelectric component may have rectangular, oval, hexagonal, triangular, and regular or irregular polygon shapes. In various implementations, the mesh plate may have a variety of different configurations. For example, in some implementations the mesh plate may have a substantially flat profile. In other implementations, the mesh plate may have a substantially domed shape, which may be concave or convex with respect to the liquid composition. In other implementations, the mesh plate may include a substantially flat portion and a domed portion. In

various implementations, the mesh plate may be made of a variety of different materials. In some implementations, the mesh plate may be made of a metal material, such as, but not limited to, stainless steel, palladium-nickel, or titanium. In other implementations, the mesh plate may be made of a polymeric material, such as, for example, a polyimide polymer. In still other implementations, the mesh plate may be made of a combination of materials.

In various implementations, the structure of the atomization assembly may vary. For example, FIGS. 4A-4F illustrate example implementations of various atomization assemblies. In particular, FIG. 4A illustrates an atomization assembly comprising a piezoelectric ring 217A affixed to and substantially surrounding a mesh plate 219A. FIG. 4B illustrates an atomization assembly comprising a mesh plate 219A sandwiched between two portions of piezoelectric ring 217A. FIG. 4C illustrates an atomization assembly comprising a piezoelectric ring 217C affixed to and substantially surrounding a mesh plate 219C, wherein at least a portion of the mesh plate 219C is curved. FIG. 4D illustrates an atomization assembly comprising a mesh plate 219D sandwiched between two portions of a piezoelectric ring 217D, wherein at least a portion of the mesh plate 219D is curved. FIG. 4E illustrates an atomization assembly comprising a piezoelectric ring 217E affixed to and substantially surrounding one side of a mesh plate 219E, wherein the other side of the mesh plate 219E includes a metal ring 221E substantially surrounding and affixed thereto. FIG. 4F illustrates an atomization assembly comprising a mesh plate 219F one side of which includes a metal ring 221F substantially surrounding and affixed thereto, the mesh plate 219F and metal ring 221F sandwiched between two portions of a piezoelectric ring 217F.

Referring back to FIG. 3, the mesh plate 219 includes a plurality of perforations. In some implementations, the perforations may be defined by circular openings in the surfaces of plate. In other implementations, the perforations may be defined by non-circular openings in the surfaces of the plate, such as, for example, oval, rectangular, triangular, or regular or irregular polygon openings. In various implementations, the perforations may be created using a variety of different methods, including, but not limited to, via a laser (e.g., a femtosecond laser) or via electroplating (e.g., lithography) or via use of high or low energy ion or electron beams. In various implementations, the shapes defined through the plate by the perforations may vary. For example, in some implementations the shapes defined through the plate by the perforations may be substantially cylindrical. In other implementations, the shapes defined through the plate by the perforations may be substantially conical (e.g., having a truncated conical shape defining smaller openings on one surface of the plate and larger openings on the opposite surface of the plate). In other implementations, the shapes defined through the plate by the perforations may be tetragonal or pyramidal. It is believed that in some implementations, substantially conical shapes may increase the performance of the mesh in atomizing the liquid composition. Although any orientation of the mesh plate may be used, in some implementations with perforations defining substantially conical shapes through the plate, the larger openings may be located proximate the surface of the liquid composition and the smaller openings may define an aerosol outlet area. In some implementations with perforations defining substantially conical shapes, the smaller openings may have a size in the inclusive range of approximately 1 micron up to approximately 10 microns, with an average size of approximately 2 microns to approximately 5 microns. In

other implementations, the smaller openings may have a size in the inclusive range of approximately several hundred nanometers up to approximately 4 microns, with an average size of approximately 2 microns to approximately 3.1 microns. In other implementations, the smaller openings may have a size in the inclusive range of approximately several hundred nanometers to approximately 2 microns, with an average size of approximately 1 micron. In some implementations, the larger openings may have a size in the inclusive range of approximately 10 microns to approximately 60 microns, with an average size of approximately 20 microns to approximately 30 microns. In other implementations, the larger openings may have a size in the inclusive range of approximately 5 microns to approximately 20 microns, with an average size of approximately 10 microns. In some implementations, the size of the perforations may be substantially uniform across the perforated portion of the plate; however, in other implementations, the size of the perforations may vary. In such a manner, the formed aerosol may have different size aerosol droplets. For example, in some implementations the perforations may be larger in one portion of the plate and smaller in another portion of the plate. Such portions may include, for example, the center of the plate and a periphery of the plate, or alternating rings that extend radially from the center of the plate.

In various implementations, the mesh plate may have any number of perforations. In some implementations, for example, a number of perforations in the mesh plate may be in the inclusive range of approximately 200 to approximately 6,000, with an average number of perforations of approximately 1,100 to approximately 2,500. In other implementations, a number of perforations in the mesh plate may be in the inclusive range of approximately 400 to approximately 1,000. In various implementations, the thickness of the piezoelectric ring and the thickness of the mesh plate may vary. For example, in some implementations the thickness of the mesh plate may be in the range of a few microns to a few millimeters. In various implementations, the overall diameter of a mesh plate may vary. For example, in some implementations the overall diameter of the mesh plate may be in the inclusive range of approximately a few millimeters to approximately 30 millimeters. In some implementations, the outer diameter of the piezoelectric ring may be larger than the overall diameter of the mesh plate. In other implementations, the outer diameter may be substantially the same size as the overall diameter of the mesh plate. In various implementations, the diameter of the perforation area may be smaller than the overall diameter of the mesh plate. For example, in some implementations the diameter of the perforated area may be in the inclusive range of approximately 1 millimeter to approximately 20 millimeters, with an average of approximately 4 millimeters to approximately 12 millimeters. In some implementations, the inner diameter of the piezoelectric ring may be larger than the diameter of the perforated area of the mesh plate. In other implementations, the inner diameter of the piezoelectric ring may be substantially the same as, or smaller than, the diameter of the perforated area of the mesh plate. In some implementations, the thickness of the piezoelectric ring may be in the inclusive range of a few hundred microns to tens of millimeters. For example, in some implementations the thickness of the piezoelectric ring may be smaller than 1 millimeter.

In various implementations, the piezoelectric ring may be made of a piezoceramic material. In general, piezoceramic materials possess piezoelectric properties (e.g., ferroelectric properties), wherein they are configured to change shape to a small extent (e.g., 1-2 microns) when exposed to an

electrical stimulus. This occurs due to a shift in the crystal structure of the piezoceramic materials (e.g., from orthorhombic to cubic, or hexagonal to cubic, etc.). With respect to a piezoceramic ring, such a change in shape results in an internal strain and therefore shrinkage of the disc that results in bending of the disk due to its rigid structure. Because the ring is affixed to the mesh plate, the bending of the ring is transferred to the mesh material. When the electric current is disconnected from the piezoelectric ring, the ring and mesh plate return to their original shape and position. As such, a continuous change of the shape and position will result in an oscillating motion that can be used as a vibration source. In various implementations, the frequency of the piezoelectric ring may be in the range of a few Hz to several MHz. For example, in some implementations the frequency of the piezoelectric ring is in the inclusive range of approximately 50 KHz to approximately 150 KHz, with an average, in one implementation, of approximately 110 KHz, and in another implementation, of approximately 113 KHz, and in another implementation, of approximately 117 KHz, in another implementation, of approximately 130 KHz, in another implementation, of approximately 150 KHz, in another implementation, of approximately 170 KHz, and in another implementation, of approximately 250 KHz. In other implementations, the frequency of the piezoelectric ring is in the inclusive range of approximately 1 MHz to approximately 5 MHz, with an average of approximately 3 MHz to approximately 3.5 MHz.

In various implementations of the present disclosure, a variety of different piezoelectric materials are possible, including natural or synthetic materials. Some non-limiting examples of natural piezoelectric materials include, for example, quartz, berlinite (AlPO_4), sucrose, rochelle salt, topaz, tourmaline-group minerals, lead titanate (PbTiO_3), and collagen. Some non-limiting examples of synthetic materials include, for example, a ($\text{La}_3\text{Ga}_5\text{SiO}_{14}$), gallium phosphate, gallium orthophosphate (GaPO_4), lithium niobate (LiNbO_3), lithium tantalate (LiTaO_3), AlN , ZnO , barium titanate (BaTiO_3), lead zirconate titanate ($\text{Pb}[\text{Zr}_x\text{Ti}_{1-x}\text{O}_3]$) (a.k.a. PZT), potassium niobate (KNbO_3), sodium tungstate (Na_2WO_3), $\text{Ba}_2\text{NaNb}_5\text{O}_{15}$, $\text{Pb}_2\text{KNb}_5\text{O}_{15}$, zinc oxide (ZnO), sodium potassium niobate ($(\text{K},\text{Na})\text{NbO}_3$) (a.k.a. NKN), bismuth ferrite (BiFeO_3), sodium niobate (NaNbO_3), barium titanate (BaTiO_3), bismuth titanate ($\text{Bi}_4\text{Ti}_3\text{O}_{12}$), sodium titanate, and sodium bismuth titanate ($\text{NaBi}(\text{TiO}_3)_2$). In other implementations, polymers exhibiting piezoelectric characteristics may be used, including, but not limited to, polyvinylidene fluoride (PVDF).

In various implementations, the mesh plate **219** of the atomization assembly **215** may be in contact with at least a portion of a liquid composition, and/or may be proximate at least a portion of a liquid composition, and/or may receive (such as via a delivery mechanism) at least a portion of a liquid composition. In such a manner, the resulting vibration of the plate generates an aerosol from the contacted liquid composition. In particular, the liquid composition is driven through the plurality of perforations resulting in the generation of a plurality of aerosol particles. Likewise, in other implementations, such as, for example, implementations in which the mesh plate is immersed in the liquid composition, vibration of the plate creates ultrasonic waves within the liquid composition that result in the formation of an aerosol at the surface of the liquid composition. As will be described in more detail below, in other implementations the liquid composition may be applied and/or transferred to the atomization assembly to create the aerosol.

Another example of an atomization assembly of one implementation is shown in FIG. 5. In particular, FIG. 5 illustrates an atomization assembly **315** that comprises a piezoelectric ring **317** that is affixed to and substantially surrounds a mesh plate **319**. As illustrated in the figure, the atomization assembly **315** is located proximate one end of a reservoir **344** containing a liquid composition **345**. The mesh plate **319** of the depicted implementation includes two portions, an outer portion **319A** that is substantially flat, and an inner portion **319B** that is domed. In the depicted implementation, inner domed portion **319B** of the mesh plate **319** is configured to interact with the liquid composition **345** and has a convex configuration with respect to the reservoir **344** (and liquid composition **345**). In the depicted implementation, the mesh plate **319** includes a plurality of perforations **360** that have a substantially conical shape. In particular, the plurality of perforations **360** include a larger end **360A**, configured to be located proximate the interface with the liquid composition **345**, and a smaller end **360B**, through which the formed aerosol passes.

In various implementations, it may be advantageous to facilitate contact (e.g., maintain or encourage contact) between the liquid composition and the atomization assembly. Along those lines, FIG. 6 illustrates a side schematic view of a portion of a reservoir **444** containing a liquid composition **445**, and an atomization assembly **415** configured to generate an aerosol from the liquid composition, according to an example implementation of the present disclosure, and FIG. 7 illustrates a perspective schematic view of a portion of the reservoir **444** containing the atomization assembly **415** (wherein the liquid composition has been removed for clarity of illustration). In the depicted implementation, the reservoir **444** has a substantially hollow spherical shape; however, in other implementations, other shapes are possible, including, for example, a substantially hollow cylindrical shape, a substantially hollow prismatic shape, a substantially hollow cuboidal shape, or any other shape configured to contain a liquid composition. Regardless of the shape of the reservoir, in various implementations the reservoir may be located in various positions within the housing of an aerosol delivery device.

In some implementations, the reservoir may be made of a polymeric material that, in further implementations, may be at least partially transparent or translucent. In some implementations, such materials, may include, but need not be limited to, polycarbonate, acrylic, polyethylene terephthalate (PET), amorphous copolyester (PETG), polyvinyl chloride (PVC), liquid silicone rubber (LSR), cyclic olefin copolymers, polyethylene (PE), ionomer resin, polypropylene (PP), fluorinated ethylene propylene (FEP), styrene methyl methacrylate (SMMA), styrene acrylonitrile resin (SAN), polystyrene, acrylonitrile butadiene styrene (ABS), and combinations thereof. In other implementations, the reservoir may be made of other material that may be at least partially transparent or translucent. Such materials may include, for example, glass or ceramic materials.

In the depicted implementation the reservoir **444** includes an opening **446** proximate a top end thereof. In various implementations, the opening **446** may be used to fill the reservoir **444** with the liquid composition **445**. In the depicted implementation, the opening **446** may be covered with a reservoir lid **448**, which, in some implementations, may include a plurality of openings and/or may be substantially gas permeable, so as to permit the formed aerosol to exit the reservoir. Although in other implementations the atomization assembly may be positioned in other locations within the reservoir, the atomization assembly **415** of the

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depicted assembly is located opposite the opening **446** of the reservoir **444** proximate the bottom thereof. In the depicted implementation, the reservoir **444** also includes an electrical connection **416** that extends to the atomization assembly **415** and electrically connects (either directly, or indirectly through one or more additional components) the atomization assembly **415** to the control component and/or the battery. In various implementations, the electrical connection **416** may extend inside and/or outside of the reservoir **444**.

In various implementations, the reservoir of the present disclosure may be configured to rotate, either freely or otherwise, such as via active control, relative to the position of the aerosol delivery device. As illustrated in the figures, the reservoir **444** of the depicted implementation is configured to rotate about one axis (axis **450**), via a pair of rotation elements **452** that are attached to the aerosol delivery device (not shown). In various implementations, the rotation elements **452** may be attached to the cartridge housing, or the control unit housing, or any other component of the aerosol delivery device that would permit the reservoir to rotate relative thereto. In various implementations, the rotation elements may comprise a variety of different components configured to permit rotation of the reservoir relative to the aerosol delivery device, including, for example, bearing elements, pins configured to rotate within corresponding detents or holes, etc. In some implementations, rotation elements **452** may facilitate electrical connection between the electrical connection **416** and the battery and/or control component.

In various implementations, rotation of the reservoir **444** of the depicted implementation is configured to facilitate contact between the liquid composition **445** and the atomization assembly **415**, independent of the position of the aerosol delivery device. In order to facilitate contact between the liquid composition **445** and the atomization assembly **415** of the depicted implementation, the bottom end of the reservoir **444** (e.g., the portion of the reservoir **444** proximate the atomization assembly **415**) of the depicted implementation is weighted relative to the top end. In various implementations, weighting of the reservoir may be accomplished in a variety of different ways. For example, in some implementations the liquid composition itself may provide sufficient weight to allow rotation of the reservoir relative to the aerosol delivery device. In other implementations, the atomization assembly may provide sufficient weight to allow rotation of the reservoir relative to the aerosol delivery device. In other implementations, a weight may be added to the reservoir proximate the atomization assembly to allow rotation of the reservoir relative to the aerosol delivery device. In still other implementations, the reservoir itself may be configured (such as, for example, by providing additional material proximate the atomization assembly) to provide sufficient weight to allow rotation of the reservoir relative to the aerosol delivery device. In some implementations, the reservoir may be configured to actively rotate relative to the aerosol delivery device. For example, in some implementations the reservoir may be actively rotated relative to the position of the aerosol delivery device, using, for example, electromagnetic means. Such implementations may include a reservoir that is configured to rotate within another part (e.g., a substantially spherical reservoir configured to rotate within another substantially spherical part) in the presence of a hydraulic fluid.

Some additional examples of reservoir configurations are shown in FIGS. **8A-8C**, which illustrate a series of top schematic views of portions of reservoirs and atomization assemblies of various aerosol delivery devices, according to

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example implementations of the present disclosure. In particular, FIG. **8A** illustrates a substantially cylindrical reservoir **544** (including an atomization assembly **515** located proximate a bottom end thereof) configured to rotate about axis **550** via rotation elements **552**, although in other implementations, the reservoir may have any shape. FIG. **8B** illustrates a substantially cuboidal reservoir **644** (that includes an atomization assembly **615** located proximate a bottom end thereof) configured to rotate about axis **650** via rotation elements **652**, although in other implementations, the reservoir may have any shape. And FIG. **8C** illustrates a substantially oval-shaped aerosol delivery device **700** having a substantially cylindrical reservoir **744** (that includes an atomization assembly **715** located proximate a bottom end thereof) configured to rotate about axis **750** via rotation elements **752**. In FIG. **8C**, the reservoir **744** is located proximate the side of the aerosol delivery device **700** closest to the opening **728** in the aerosol delivery device through the aerosol is delivered to a user. By forming the aerosol delivery device as such and by positioning the opening **728** of the mouthpiece portion in this location, the reservoir **744** of the depicted implementation may facilitate contact between the liquid composition in the reservoir **744** and the atomization assembly **715** independent of the position of the aerosol delivery device and via rotation of the reservoir **744** about the axis **750**.

It should be noted that while in some implementations, an aerosol delivery device of the present disclosure may comprise separate cartridge and control unit components, wherein the reservoir may or may not be refillable, in other implementations an aerosol delivery device of the present disclosure may comprise a unitary body, wherein the reservoir may or may not be refillable. For example, FIG. **9** illustrates an aerosol delivery device, according to another example implementation of the present disclosure. In particular, FIG. **9** illustrates a perspective schematic view of an aerosol delivery device **800**. As depicted in the figure, the aerosol delivery device **800** does not include separate cartridge and control units, but rather, those components are combined in a unitary body having a common housing **805**. As depicted in the figure, the housing **805** includes a mouthpiece portion having an opening **828**. The aerosol delivery device **800** of the depicted implementation also includes a reservoir **844** having a substantially spherical shape. In the depicted implementation, the reservoir **844** includes an opening **846** proximate a top end thereof. In various implementations, the opening **846** may be used to fill the reservoir **844** with the liquid composition. In the depicted implementation, the opening **846** may be covered with a reservoir lid **848**. Although in other implementations the atomization assembly may be positioned in other locations within the reservoir, the atomization assembly **815** of the depicted assembly is located opposite the opening **846** of the reservoir **844** proximate the bottom thereof. In the depicted implementation, the reservoir **844** also includes an electrical connection (not shown) that connects the atomization assembly **815** (either directly, or indirectly through one or more additional components) to the control component and/or the battery. In various implementations, the electrical connection may extend inside and/or outside of the reservoir **844**. In some implementations, the rotation elements **852** may facilitate electrical connection between the atomization assembly **815** and the battery and/or control component.

In various implementations, the reservoir of the present disclosure may be configured to rotate, either freely or otherwise, such as via active control, relative to the position

of the aerosol delivery device. As illustrated in the figures, the reservoir **844** of the depicted implementation is configured to rotate about one axis (axis **850**), via a pair of rotation elements **852** that are attached to the aerosol delivery device **800**. In various implementations, the rotation elements **852** may be attached to the housing **805** or any component of the aerosol delivery device that would permit the reservoir to rotate relative thereto. In various implementations, the rotation elements may comprise a variety of different components configured to permit rotation of the reservoir relative to the aerosol delivery device, including, for example, bearing elements, pins configured to rotate within corresponding detents or holes, etc.

In various implementations, the rotation of the reservoir **844** of the depicted implementation is configured to facilitate contact between the liquid composition and the atomization assembly **815**, independent of the position of the aerosol delivery device. In order to facilitate contact between the liquid composition and the atomization assembly **815** of the depicted implementation, the bottom end of the reservoir **844** (e.g., the portion of the reservoir **844** proximate the atomization assembly **815**) of the depicted implementation is weighted relative to the top end. In various implementations, weighting of the reservoir may be accomplished in a variety of different ways. For example, in some implementations the liquid composition itself may provide sufficient weight to allow rotation of the reservoir relative to the aerosol delivery device. In other implementations, the atomization assembly may provide sufficient weight to allow rotation of the reservoir relative to the aerosol delivery device. In other implementations, a weight may be added to the reservoir proximate the atomization assembly to allow rotation of the reservoir relative to the aerosol delivery device. In still other implementations, the reservoir itself may be configured (such as, for example, by providing additional material proximate the atomization assembly) to provide sufficient weight to allow rotation of the reservoir relative to the aerosol delivery device. In some implementations, the reservoir may be configured to actively rotate relative to the aerosol delivery device. For example, in some implementations the reservoir may be actively rotated relative to the position of the aerosol delivery device, using, for example, electromagnetic components. Such implementations may include a reservoir that is configured to rotate within another part (e.g., a substantially spherical reservoir configured to rotate within another substantially spherical part) in the presence of a hydraulic fluid.

Although in some implementations rotation of a reservoir may occur via rotation about one axis relative to the aerosol delivery device, in other implementations rotation of the reservoir may occur about two axes relative to the aerosol delivery device. A schematic example of such an implementation is illustrated in FIG. **10**. In particular, FIG. **10** illustrates a top cross-section schematic view of a portion of a reservoir **944** and atomization assembly **915** of an aerosol delivery device, according to another example implementation of the present disclosure. Although in other implementations the atomization assembly may be positioned in other locations within the reservoir, the atomization assembly **915** of the depicted assembly is located proximate the bottom of the reservoir **944**. In the depicted implementation, the reservoir **944** also includes an electrical connection (not shown) that connects the atomization assembly **915** (either directly, or indirectly through one or more additional components) to the control component and/or the battery. In various implementations, the electrical connection may extend inside and/or outside of the reservoir **944**. In some implementa-

tions, the rotation elements may facilitate electrical connection between the atomization assembly **815** and the battery and/or control component.

In various implementations, the reservoir of the present disclosure may be configured to rotate, either freely or otherwise (such as via active control) relative to the position of the aerosol delivery device. As illustrated in the figure, the implementation of FIG. **10** includes a mid-frame **960** configured to rotate relative to the housing **905** of the aerosol delivery device about a first axis (axis **950A**) via a pair of first rotation elements **952A**. In various implementations, the first rotation elements **952A** may be attached to the housing **905** or any component of the aerosol delivery device that would permit the mid-frame **960** to rotate relative thereto. In various implementations, the first rotation elements may comprise a variety of different components configured to permit rotation of the mid-frame relative to the aerosol delivery device, including, for example, bearing elements, pins configured to rotate within corresponding detents or holes, etc.

Further, in the depicted implementation the reservoir **944** is configured to rotate relative to the mid-frame **960** about a second axis (axis **950B**) via a pair of second rotation elements **952B**. In the depicted implementation, the second axis **950B** is substantially perpendicular to the first axis **950A**. In various implementations, the second rotation elements may comprise a variety of different components configured to permit rotation of the mid-frame relative to the aerosol delivery device, including, for example, bearing elements, pins configured to rotate within corresponding detents or holes, etc. In various implementations, the second rotation elements may be the same or different than the first rotation elements. In the depicted implementation, the reservoir **944** has a substantially hollow spherical shape; however, in other implementations, other shapes are possible, including, for example, a substantially hollow cylindrical shape, a substantially hollow prismatic shape, a substantially hollow cuboidal shape, or any other shape configured to contain a liquid composition. Regardless of the shape of the reservoir, in various implementations the reservoir may be located in various positions within the housing of an aerosol delivery device.

In various implementations, rotation of the reservoir **944** of the depicted implementation is configured to facilitate contact between the liquid composition and the atomization assembly **915**, independent of the position of the aerosol delivery device. In order to facilitate contact between the liquid composition and the atomization assembly **915** of the depicted implementation, the bottom end of the reservoir **944** (e.g., the portion of the reservoir **944** proximate the atomization assembly **915**) of the depicted implementation is weighted relative to the top end. In various implementations, weighting of the reservoir may be accomplished in a variety of different ways. For example, in some implementations the liquid composition itself may provide sufficient weight to allow rotation of the reservoir relative to the aerosol delivery device. In other implementations, the atomization assembly may provide sufficient weight to allow rotation of the reservoir relative to the aerosol delivery device. In other implementations, a weight may be added to the reservoir proximate the atomization assembly to allow rotation of the reservoir relative to the aerosol delivery device. In still other implementations, the reservoir itself may be configured (such as, for example, by providing additional material proximate the atomization assembly) to provide sufficient weight to allow rotation of the reservoir relative to the aerosol delivery device. In some implementa-

tations, the reservoir may be configured to actively rotate relative to the aerosol delivery device. For example, in some implementations the reservoir may be actively rotated relative to the position of the aerosol delivery device, using, for example, electromagnetic means. Such implementations may include a reservoir that is configured to rotate within another part (e.g., a substantially spherical reservoir configured to rotate within another substantially spherical part) in the presence of a hydraulic fluid.

Another implementation of the present disclosure is shown in FIG. 11. In particular, FIG. 11 illustrates a side schematic view of a portion of a reservoir 1044 configured to hold a liquid composition (not shown) and an atomization assembly 1015 configured to generate an aerosol from the liquid composition, according to another example implementation of the present disclosure. In the depicted implementation, the reservoir 1044 and atomization assembly 1015 are configured to be used in conjunction with an aerosol delivery device. In the depicted implementation, the atomization assembly 1015 comprises a vibrating mesh assembly that includes a piezoelectric disc or ring 1017 affixed to and substantially surrounding a mesh plate 1019. In various implementations, an electrical connection (not shown) connects the atomization assembly 1015 to the control component and/or battery of the aerosol delivery device. In such a manner, the atomization assembly 1015 of the depicted implementation may be energized by the battery and/or control component so as to vibrate the mesh plate 1019 at a relatively high rate. The resulting vibration of the plate generates an aerosol from the contacted liquid composition. Further reference is made to the discussion of vibrating mesh assembly components and variations described above.

In the depicted implementation, the atomization assembly 1015 further includes a perforated gate 1062 located proximate the mesh plate 1019, between the liquid composition and the mesh plate 1019. In the depicted implementation, the perforated gate 1062 is substantially parallel to the mesh plate 1019 such that a small chamber 1064 is defined between the mesh plate 1019 and the perforated gate 1062. In various implementations, the perforated gate 1062 defines a plurality of openings 1066 configured to permit the liquid composition to pass therethrough. In the depicted implementation, the openings 1066 are configured to facilitate passage of the liquid composition from the reservoir 1044 into the small chamber 1064 between the mesh plate 1019 and the perforated gate 1062.

As illustrated in the figure, each of the openings 1066 of the depicted implementation has a truncated conical shape (see detail), wherein the small end of the opening is closest to the mesh plate 1019. In such a manner, the openings 1066 may act similar to a one directional valve wherein the liquid composition is permitted to pass through the perforated gate 1062 into the chamber 1064 in one direction and is inhibited from traveling back through the perforated gate 1062 in the opposite direction. It should be noted that in other implementations, the openings may have other configurations and/or other shapes. For example, in other implementations, the plurality of openings may have tetragonal, pyramidal, or cylindrical shapes. In some implementations, at least a portion of the perforated gate 1062 may include one or more coating materials. For example, in the depicted implementation the side of the perforated gate 1062 closest to the chamber 1064 includes a hydrophobic/oleophobic coating. Because of the configuration of the reservoir 1044 and

atomization assembly 1015, the depicted implementation enables use of the aerosol delivery device at various angles and orientations.

Another implementation of the present disclosure is shown in FIG. 12. In particular, FIG. 12 illustrates a side schematic view of a portion of a reservoir 1144 configured to hold a liquid composition (not shown) and an atomization assembly 1115 configured to generate an aerosol from the liquid composition, according to another example implementation of the present disclosure. In the depicted implementation, the reservoir 1144 and atomization assembly 1115 are configured to be used in conjunction with an aerosol delivery device. In the depicted implementation, the atomization assembly 1115 comprises a vibrating mesh assembly that includes a piezoelectric disc or ring 1117 affixed to and substantially surrounding a mesh plate 1119. In various implementations, an electrical connection (not shown) connects the atomization assembly 1115 to the control component and/or battery of the aerosol delivery device. In such a manner, the atomization assembly 1115 of the depicted implementation may be energized by the battery and/or control component so as to vibrate the mesh plate 1119 at a relatively high rate. The resulting vibration of the plate generates an aerosol from the contacted liquid composition. Further reference is made to the discussion of vibrating mesh assembly components and variations described above.

The depicted implementation further includes a liquid transport element 1168, one end of which is located proximate the mesh plate 1119. In the depicted implementation, the liquid transport element 1168 is located between the liquid composition in the reservoir 1144 and the mesh plate 1119. In some implementations, the liquid transport element may take up a portion, or substantially all, of the reservoir. In various implementations, the liquid transport element may have one layer, or multiple layers, and may be made of a single material or multiple materials. In various implementations, the liquid transport element may be any shape and may be a porous, semi-porous, or non-porous absorbent/adsorbent material. In other implementations, there may be a second liquid transport element located between the first liquid transport element and the liquid reservoir, the second liquid transport element being configured to transfer liquid from the liquid reservoir to the first liquid transport element. In such a manner, the first liquid transport element may not be in direct contact with the liquid in the liquid reservoir. In various implementations, the second liquid transport element may be made of the same material or a different material than the first liquid transport element and may have a shape that is the same or differs from that of the first liquid transport element.

For example, in some implementations the liquid transport element may be made of fibrous materials (e.g., organic cotton, cellulose acetate, regenerated cellulose fabrics, glass fibers), polymers, silk, particles, porous ceramics (e.g., alumina, silica, zirconia, SiC, SiN, AlN, etc.), porous metals, porous carbon, graphite, porous glass, sintered glass beads, sintered ceramic beads, capillary tubes, porous polymers, or the like. In some implementations, the liquid transport element may be any material that contains an open pore network (i.e., a plurality of pores that are interconnected so that fluid may flow from one pore to another in a plurality of direction through the element). The pores can be nanopores, micropores, macropores or combinations thereof. As further discussed herein, some implementations of the present disclosure may particularly relate to the use of non-fibrous transport elements. As such, fibrous transport ele-

ments may be expressly excluded. Alternatively, combinations of fibrous transport elements and non-fibrous transport elements may be utilized. In some embodiments, the liquid transport element may be a substantially solid non-porous material, such as a polymer or dense ceramic or metals, configured to channel liquid through apertures or slots while not necessarily relying upon wicking through capillary action. Such a solid body may be used in combination with a porous absorptive pad. The absorptive pad may be formed of silica-based fibers, organic cotton, rayon fibers, cellulose acetate, regenerated cellulose fabrics, highly porous ceramic or metal mesh, etc. Some representative types of substrates, reservoirs or other components for supporting the aerosol precursor are described in U.S. Pat. No. 8,528,569 to Newton; U.S. Pat. App. Pub. Nos. 2014/0261487 to Chapman et al. and 2014/0059780 to Davis et al.; and U.S. Pat. App. Pub. No. 2015/0216232 to Bless et al.; which are incorporated herein by reference in their entireties. Additionally, various wicking materials, and the configuration and operation of those wicking materials within certain types of electronic cigarettes, are set forth in U.S. Pat. No. 8,910,640 to Sears et al.; which is incorporated herein by reference in its entirety. In some implementations, the liquid transport element may be formed partially or completely from a porous monolith, such as a porous ceramic, a porous glass, or the like. Example monolithic materials that may be suitable for use according to embodiments of the present disclosure are described, for example, in U.S. Pat. App. Pub. No. 2017/0188626 to Davis et al., and U.S. Pat. App. Pub. No. 2014/0123989 to LaMothe, the disclosures of which are incorporated herein by reference in their entireties. In some implementations, the porous monolith may form a substantially solid wick.

In various implementations, an end of the liquid transport element is configured to be placed proximate the mesh plate and between the mesh plate and liquid composition in the reservoir so that the liquid transport element acts as a secondary reservoir that absorbs or adsorbs the liquid from the reservoir so that at least a portion of the liquid composition is delivered to the mesh plate, even if there is no longer liquid in the reservoir. In such a manner, the liquid transport element is configured to facilitate delivery of the liquid composition to the atomization assembly.

Another implementation of the present disclosure is shown in FIG. 13. In particular, FIG. 13 illustrates a side schematic view of a portion of a reservoir **1244** configured to hold a liquid composition (not shown) and an atomization assembly **1215** configured to generate an aerosol from the liquid composition, according to another example implementation of the present disclosure. In the depicted implementation, the reservoir **1244** and atomization assembly **1215** are configured to be used in conjunction with an aerosol delivery device. In the depicted implementation, the atomization assembly **1215** comprises a vibrating mesh assembly that includes a piezoelectric disc or ring **1217** affixed to and substantially surrounding a mesh plate **1219**. In various implementations, an electrical connection (not shown) connects the atomization assembly **1215** to the control component and/or battery of the aerosol delivery device. In such a manner, the atomization assembly **1215** of the depicted implementation may be energized by the battery and/or control component so as to vibrate the mesh plate **1219** at a relatively high rate. Because at least a portion of the liquid composition is delivered to the mesh plate **1219**, the resulting vibration of the plate generates an aerosol from the contacted liquid composition. Further reference is made

to the discussion of vibrating mesh assembly components and variations described above.

In the depicted implementation, the atomization assembly **1215** further includes a micropump assembly **1260**. In various implementations, the micropump assembly **1260** is configured to transfer a portion of the liquid composition from the reservoir **1244** to the mesh plate **1219**. In some implementations, delivery of the liquid composition to the mesh plate **1219** may occur automatically. In other implementations, delivery of the liquid composition to the mesh plate **1219** may occur on demand, such as, for example, via control from the control component. In some implementations, the micropump assembly may apply liquid composition droplets directly onto the mesh plate. In other implementations, the micropump assembly may spray (e.g., via one or more nozzles) liquid composition droplets onto the mesh plate. In still other implementations, the micropump assembly may apply or spray liquid composition droplets onto a liquid transport element, which may deliver at least a portion of the liquid composition to the mesh plate, as described with respect to FIG. 12 above. In various implementations, the micropump assembly may comprise any mechanical or non-mechanical pump configured to transfer liquid from one location to another. Non-limiting examples of micropump assemblies include dispensing mechanisms (such as, for example, shape-memory dispensing mechanisms), diaphragm devices, peristaltic devices, or lab-on-a-chip or lab-on-a-disk microfluidic systems, etc. Such devices may be configured to automatically transfer the liquid or may initial transfer due to one or more stimuli, such as manual force or pressure, etc. Some examples of shape-memory dispensing mechanisms are described in U.S. Pat. No. 10,080,388 to Sebastian et al., which is incorporated herein by reference in its entirety.

Another implementation of the present disclosure is shown in FIG. 14. In particular, FIG. 14 illustrates a side schematic view of a portion of a reservoir **1344** configured to hold a liquid composition (not shown) and an atomization assembly **1315** configured to generate an aerosol from the liquid composition, according to another example implementation of the present disclosure. In the depicted implementation, the reservoir **1344** and atomization assembly **1315** are configured to be used in conjunction with an aerosol delivery device. In the depicted implementation, the atomization assembly **1315** comprises a vibrating mesh assembly that includes a piezoelectric disc or ring **1317** affixed to and substantially surrounding a mesh plate **1319**. In various implementations, an electrical connection (not shown) connects the atomization assembly **1315** to the control component and/or battery of the aerosol delivery device. In such a manner, the atomization assembly **1315** of the depicted implementation may be energized by the battery and/or control component so as to vibrate the mesh plate **1319** at a relatively high rate. Because at least a portion of the liquid composition is delivered to the mesh plate **1319**, the resulting vibration of the plate generates an aerosol from the contacted liquid composition. Further reference is made to the discussion of vibrating mesh assembly components and variations described above.

In the depicted implementation, the atomization assembly **1315** further includes a micro blower assembly **1362**. In various implementations, the micro blower assembly **1362** is configured to propel a portion of the liquid composition (e.g., in the form of small particles or a small stream) from the reservoir **1344** to the surface of the mesh plate **1319**. In some implementations, the micro blower assembly may comprise a micro compressor. In some implementations, the

micro blower assembly may utilize a pressurized gas (e.g., air, carbon dioxide (CO₂), nitrogen (N₂), etc.) to aid in propelling the liquid composition to the surface of the mesh plate. In some implementations, the micro blower assembly may include one or more nozzles.

Other implementations of the present disclosure are shown in FIGS. 15A and 15B. In particular, FIGS. 15A and 15B illustrate side schematic views of a portion of a reservoir 1444A, 1444B configured to hold a liquid composition 1445A, 1445B and an atomization assembly 1415A, 1415B configured to generate an aerosol from the liquid composition, according to other example implementations of the present disclosure. In the depicted implementations, the reservoirs 1444A, 1444B and atomization assemblies 1415A, 1415B are configured to be used in conjunction with an aerosol delivery device. In the depicted implementation, each atomization assembly 1415A, 1415B comprises a vibrating mesh assembly that includes a piezoelectric disc or ring 1417A, 1417B affixed to and substantially surrounding a mesh plate 1419A, 1419B. In various implementations, an electrical connection (not shown) connects each atomization assembly 1415A, 1415B to the control component and/or battery of the aerosol delivery device. In such a manner, the atomization assemblies 1415A, 1415B of the depicted implementations may be energized by the battery and/or control component so as to vibrate the respective mesh plates 1419A, 1419B at a relatively high rate. The resulting vibration of the plates generates a respective aerosol from the contacted liquid composition. Further reference is made to the discussion of vibrating mesh assembly components and variations described above.

In the depicted implementation, the reservoirs 1444A, 1444B comprise substantially U-shaped tubes and include respective first reservoir sections 1447A, 1447B, second reservoir sections 1449A, 1449B, and third reservoir sections 1451A, 1451B. In the depicted implementations, the first reservoir sections 1447A, 1447B and the third reservoir sections 1451A, 1451B are substantially straight, and the second reservoir sections 1449A, 1449B, which connect the first reservoir sections 1447A, 1447B to the third reservoir sections 1451A, 1451B, are substantially curved. In the implementation depicted in FIG. 15A, the diameters of the first reservoir section 1447A, the second reservoir section 1449A, and the third reservoir section 1451A are substantially the same. In other implementations, however, the diameters may differ. For example, in the implementation depicted in FIG. 15B, the diameters of the second reservoir section 1449B and the third reservoir section 1451B are substantially the same and smaller than the diameter of the first reservoir section 1447B.

In each of the depicted implementations, the reservoirs 1444A, 1444B include respective plummets 1464A, 1464B, which exert a downward force on the liquid composition in the first reservoir sections 1447A, 1447B. In various implementations, the plummet may comprise any device configured to increase the pressure on the liquid composition, including, for example, weighted discs or balls. In such a manner, even as the level of liquid composition in the reservoirs 1444A, 1444B decreases, contact between the liquid composition and the mesh plates 1419A, 1419B is maintained.

Another implementation of the present disclosure is shown in FIG. 16. In particular, FIG. 16 illustrates a side schematic view of a portion of a reservoir 1544 configured to hold a liquid composition 1545, and an atomization assembly 1515 configured to generate an aerosol from the liquid composition, according to another example imple-

mentation of the present disclosure. In the depicted implementations, the reservoir 1544 and atomization assembly 1515 are configured to be used in conjunction with an aerosol delivery device such as those described above. In the depicted implementation, the atomization assembly 1515 comprises a vibrating mesh assembly that includes a piezoelectric disc or ring 1517 affixed to and substantially surrounding a mesh plate 1519. In various implementations, an electrical connection (not shown) connects the atomization assembly 1515 to the control component and/or battery of the aerosol delivery device. In such a manner, the atomization assembly 1515 of the depicted implementations may be energized by the battery and/or control component so as to vibrate the mesh plate 1519 at a relatively high rate. The resulting vibration of the plate generates an aerosol from the contacted liquid composition. Further reference is made to the discussion of vibrating mesh assembly components and variations described above.

In the depicted implementation, the reservoir 1544 comprises a substantially U-shaped tube and includes a first reservoir section 1547, a second reservoir section 1549, and a third reservoir section 1551. In the depicted implementations, the first reservoir section 1547 and the third reservoir section 1551 are substantially straight, and the second reservoir section 1549, which connects the first reservoir section 1547 to the third reservoir section 1551, is substantially curved. In the implementation depicted in FIG. 16, the diameters of the first reservoir section 1547, the second reservoir section 1549, and the third reservoir section 1551 are substantially the same. In other implementations, however, the diameters may differ.

In the depicted implementation, the reservoir 1544 includes a plummet 1564, which exerts a downward force on the liquid composition in the first reservoir sections 1547. In various implementations, the plummet may comprise any device configured to increase the pressure on the liquid composition, including, for example, weighted discs or balls. In such a manner, even as the level of liquid composition in the reservoir 1544 decreases, contact between the liquid composition and the mesh plates 1519 is maintained. Further, in the depicted implementation the U-shaped reservoir is angled with respect to a longitudinal axis 1550 of the aerosol delivery device body 1505 such that contact between the liquid composition and the mesh plate 1519 is maintained. In addition, in the depicted implementation the mouthpiece portion 1509 (which defines the opening 1528 through which the aerosol exits) is positioned on one side of the aerosol delivery device body 1505 so as to encourage use of the device in particular orientations.

Another implementation of the present disclosure is shown in FIG. 17. In particular, FIG. 17 illustrates a side schematic view of a portion of a reservoir 1644 configured to hold a liquid composition 1645, and an atomization assembly 1615 configured to generate an aerosol from the liquid composition, according to another example implementation of the present disclosure. In the depicted implementation, the reservoir 1644 and atomization assembly 1615 are configured to be used in conjunction with an aerosol delivery device. In the depicted implementation, the atomization assembly 1615 comprises a vibrating mesh assembly that includes a piezoelectric disc or ring 1617 affixed to and substantially surrounding a mesh plate 1619. In various implementations, an electrical connection (not shown) connects the atomization assembly 1615 to the control component and/or battery of the aerosol delivery device. In such a manner, the atomization assembly 1615 of the depicted implementations may be energized by the

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battery and/or control component so as to vibrate the mesh plate 1619 at a relatively high rate. The resulting vibration of the plate generates an aerosol from the contacted liquid composition. Further reference is made to the discussion of vibrating mesh assembly components and variations described above.

In the depicted implementation, the reservoir 1644 comprises a substantially U-shaped tube and includes a first reservoir section 1647, a second reservoir section 1649, and a third reservoir section 1651. In the depicted implementations, the first reservoir section 1647 and the third reservoir section 1651 are substantially straight, and the second reservoir section 1619, which connects the first reservoir section 1647 to the third reservoir section 1651, is substantially curved. In the implementation depicted in FIG. 17, the diameters of the first reservoir section 1647, the second reservoir section 1649, and the third reservoir section 1651 are substantially the same. In other implementations, however, the diameters may differ.

In the depicted implementation, the reservoir 1644 includes a plummet 1664, which exerts a downward force on the liquid composition in the first reservoir sections 1647. As noted above, in various implementations the plummet may comprise any device configured to increase the pressure on the liquid composition. In some implementations, such as the depicted implementation, the plummet may further include an active component configured to actively apply pressure to the liquid composition in the reservoir. In various implementations, the active component may be any device configured to actively exert a force on the liquid composition in the reservoir. In the depicted implementation, the active component comprises a spring 1665 that exerts an additional force on the plummet 1664. In other implementations, the active component may comprise an electromechanical device, such as, for example, an actuator or piston, that, in some implementations, may include feedback control. In various implementations, the active component may be selected based on the properties of the liquid composition (e.g., viscosity, density, etc.) and/or the size of the mesh holes and/or the frequency of vibration of the vibrating assembly. It should also be noted that in other implementations, a reservoir having an active component may be substantially cylindrical in shape.

Although in some implementations of the present disclosure a cartridge and a control unit may be provided together as a complete aerosol delivery device generally, these components may be provided separately. For example, the present disclosure also encompasses a disposable unit for use with a reusable unit. In specific implementations, such a disposable unit (which may be a cartridge as illustrated in the appended figures) can be configured to engage a reusable unit (which may be a control unit as illustrated in the appended figures). In still other configurations, a cartridge may comprise a reusable unit and a control unit may comprise a disposable unit.

Although some figures described herein illustrate a cartridge and a control unit in a working relationship, it is understood that the cartridge and the control unit may exist as individual components. Accordingly, any discussion otherwise provided herein in relation to the components in combination also should be understood as applying to the control unit and the cartridge as individual and separate components.

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In another aspect, the present disclosure may be directed to kits that provide a variety of components as described herein. For example, a kit may comprise a control unit with one or more cartridges. A kit may further comprise a control unit with one or more charging components. A kit may further comprise a control unit with one or more batteries. A kit may further comprise a control unit with one or more cartridges and one or more charging components and/or one or more batteries. In further implementations, a kit may comprise a plurality of cartridges. A kit may further comprise a plurality of cartridges and one or more batteries and/or one or more charging components. In the above implementations, the cartridges or the control units may be provided with a heating member inclusive thereto. The inventive kits may further include a case (or other packaging, carrying, or storage component) that accommodates one or more of the further kit components. The case could be a reusable hard or soft container. Further, the case could be simply a box or other packaging structure.

Many modifications and other implementations of the disclosure will come to mind to one skilled in the art to which this disclosure pertains having the benefit of the teachings presented in the foregoing descriptions and the associated drawings. Therefore, it is to be understood that the disclosure is not to be limited to the specific embodiments disclosed herein and that modifications and other embodiments are intended to be included within the scope of the appended claims. Although specific terms are employed herein, they are used in a generic and descriptive sense only and not for purposes of limitation.

The invention claimed is:

1. An aerosol delivery device comprising:
 - a housing defining an outer wall, and further including a power source and a control component;
 - a mouthpiece portion;
 - a tank portion that includes a reservoir configured to contain a liquid composition; and
 - an atomization assembly configured to vaporize the liquid composition to generate an aerosol, wherein the atomization assembly comprises a vibrating assembly that includes a mesh plate, and further including a liquid transport element, one end of which is located proximate the mesh plate, wherein the liquid transport element comprises multiple layers of a single material.
2. The aerosol delivery device of claim 1, wherein the liquid transport element comprises at least one of a polymer material, a polymer fiber material, a cotton material, a silk material, a silica fiber material, a particulate material, a synthetic fiber material, a natural fiber material, and a ceramic material.
3. The aerosol delivery device of claim 1, wherein the vibrating assembly further includes a piezoelectric component affixed to and substantially surrounding the mesh plate.
4. The aerosol delivery device of claim 1, wherein the mesh plate is substantially flat.
5. The aerosol delivery device of claim 1, wherein at least a portion of the mesh plate is convex with respect to the reservoir.

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